Class 17: Semantic Communication

From Delivering Bits to Delivering Meaning and Tasks

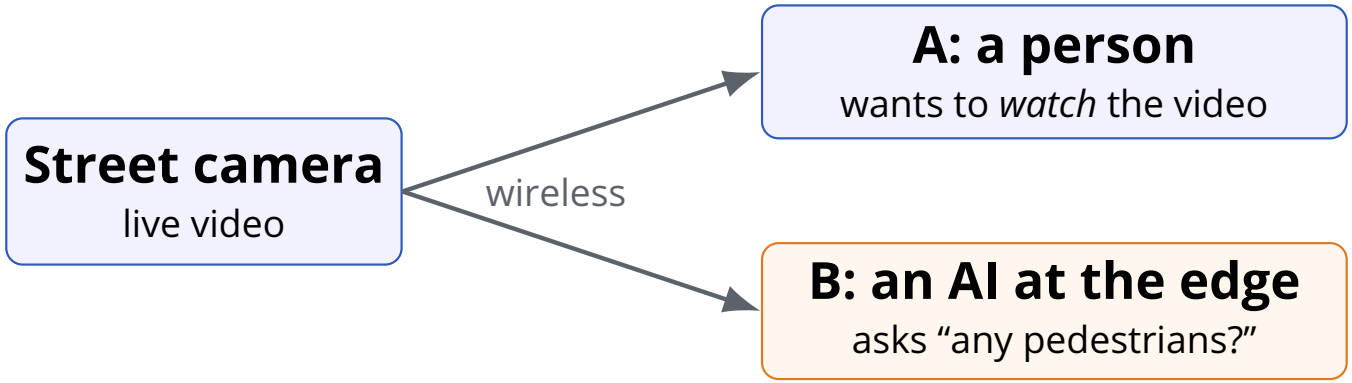
Xin Liu

Florida State University
xliu15@fsu.edu

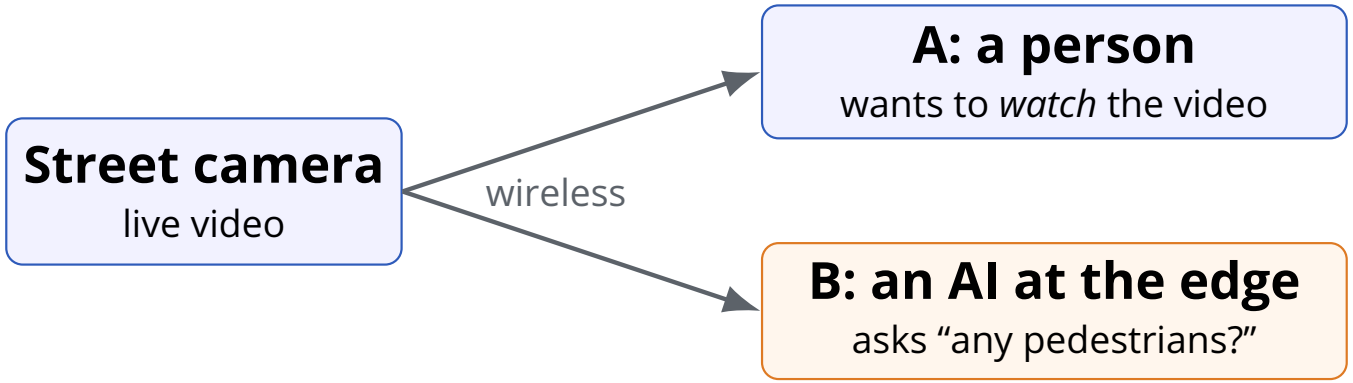
CIS 4930/5930 Future Edge Networks

xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks

One video stream, two receivers

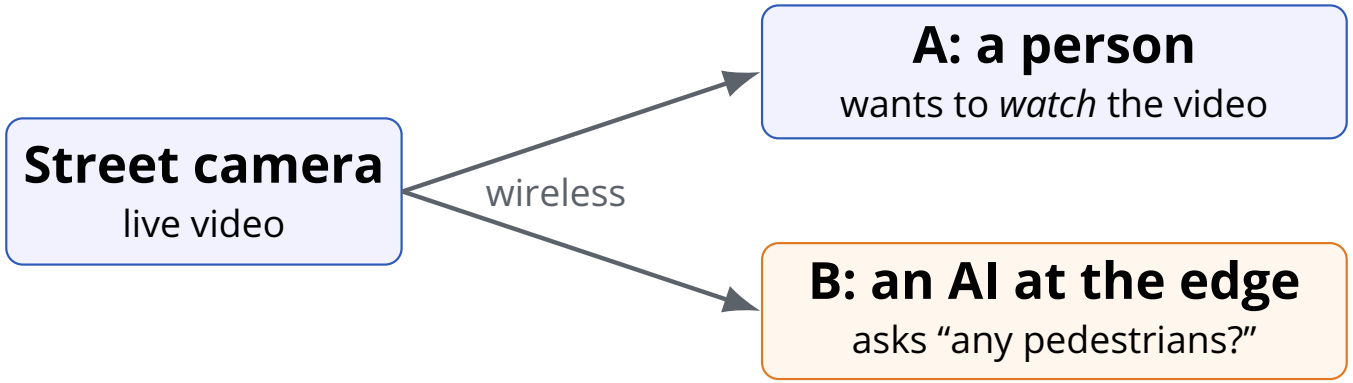


One video stream, two receivers



Do A and B need the same data?

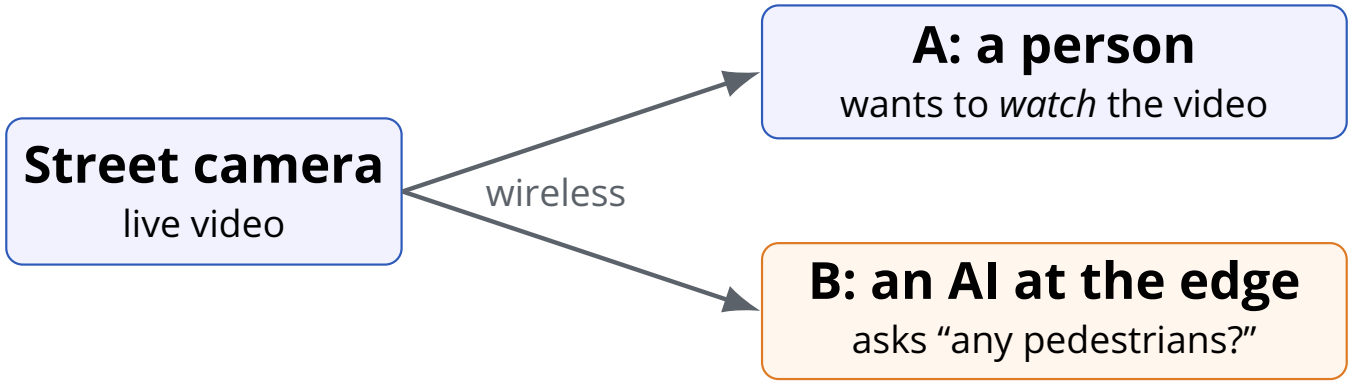
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Do A and B need the same data?

1. Today: copy every bit of every frame

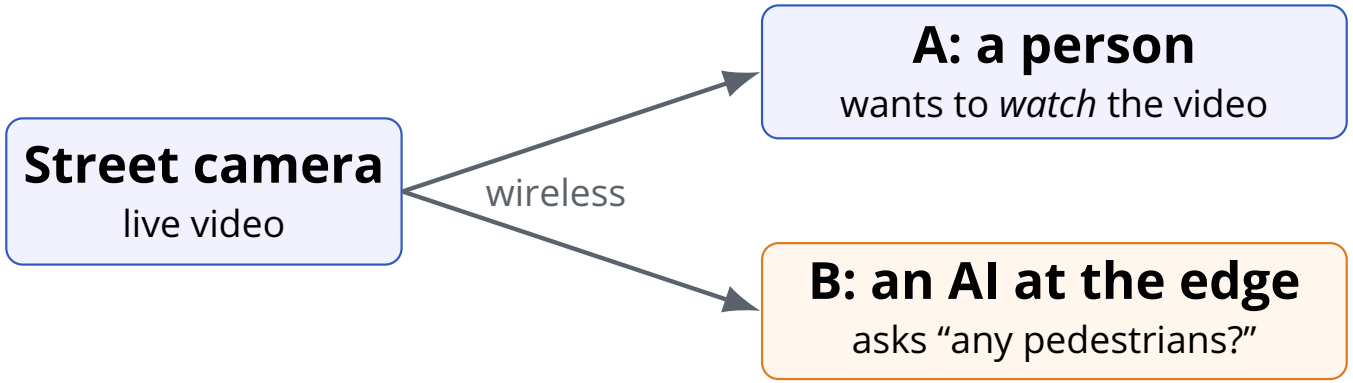
One video stream, two receivers



Do A and B need the same data?

1. Today: copy every bit of every frame
2. Semantic communication: send the **meaning**

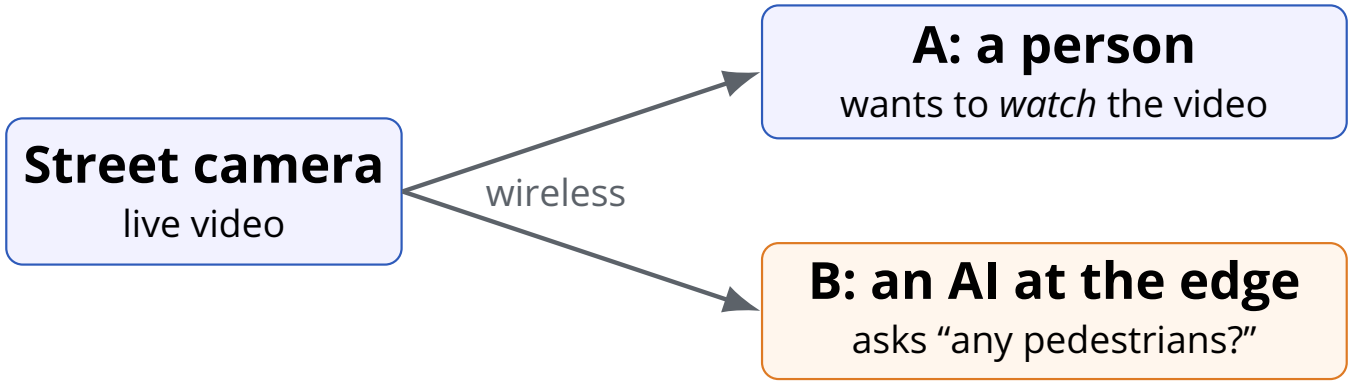
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Do A and B need the same data?

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3. Version 1: **rebuild** the video

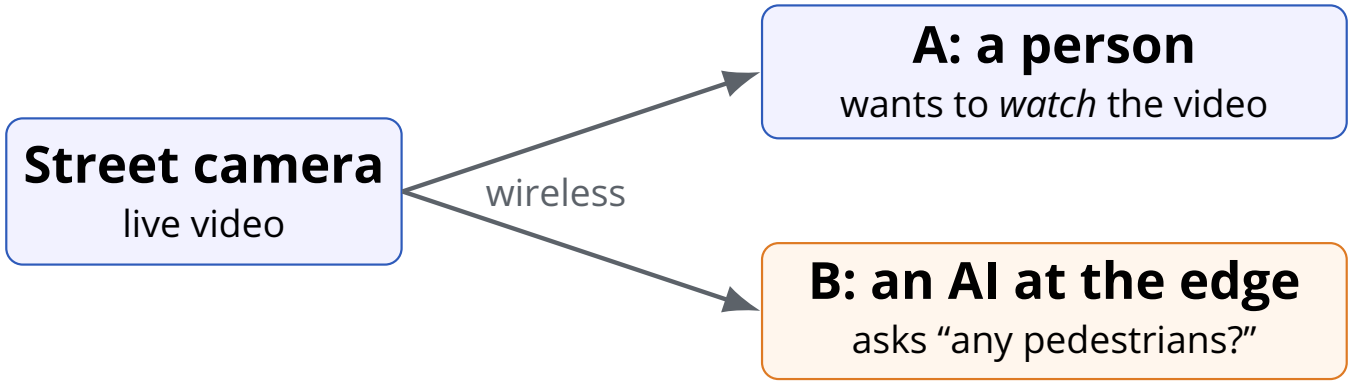
One video stream, two receivers



Do A and B need the same data?

1. Today: copy every bit of every frame
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3. Version 1: **rebuild** the video
4. Version 2: just get the **task** done — no video rebuilt

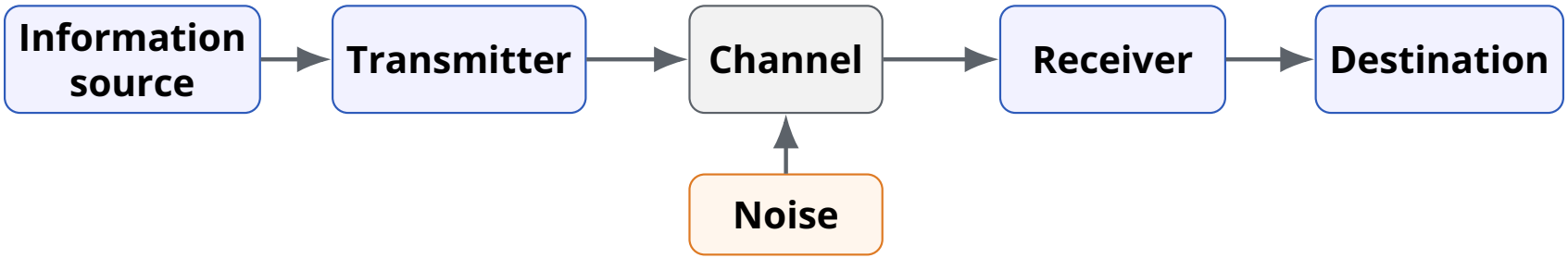
One video stream, two receivers



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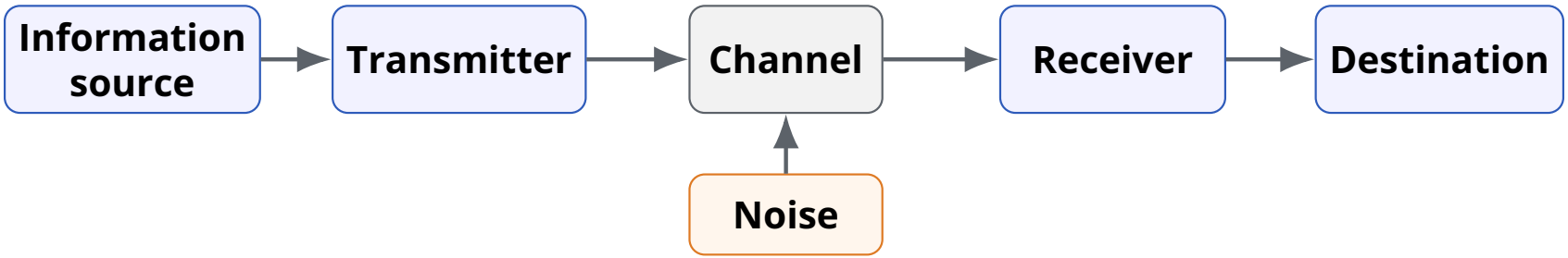
1. Today: copy every bit of every frame
2. Semantic communication: send the **meaning**
3. Version 1: **rebuild** the video
4. Version 2: just get the **task** done — no video rebuilt
5. Version 3: keep today’s pipe, spend bits by **meaning**

Shannon 1948: copy the message



C. E. Shannon, “A Mathematical Theory of Communication,” Bell Syst. Tech. J., 1948

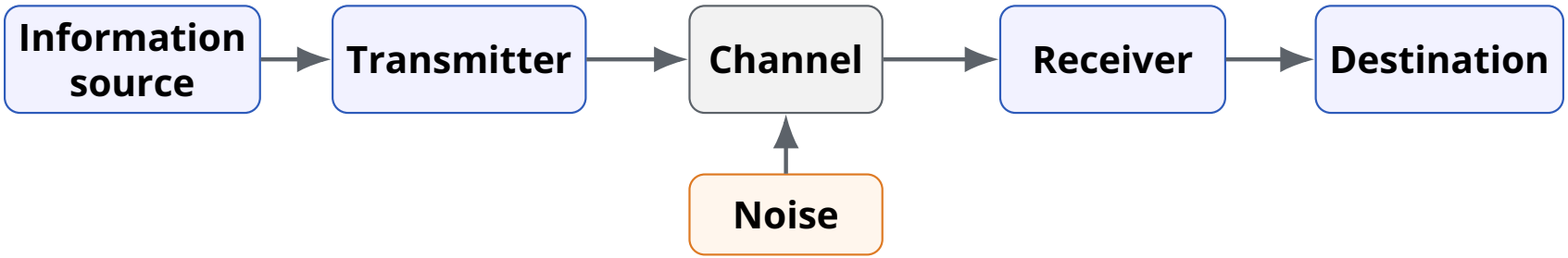
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Goal: an **exact copy** at the other end.

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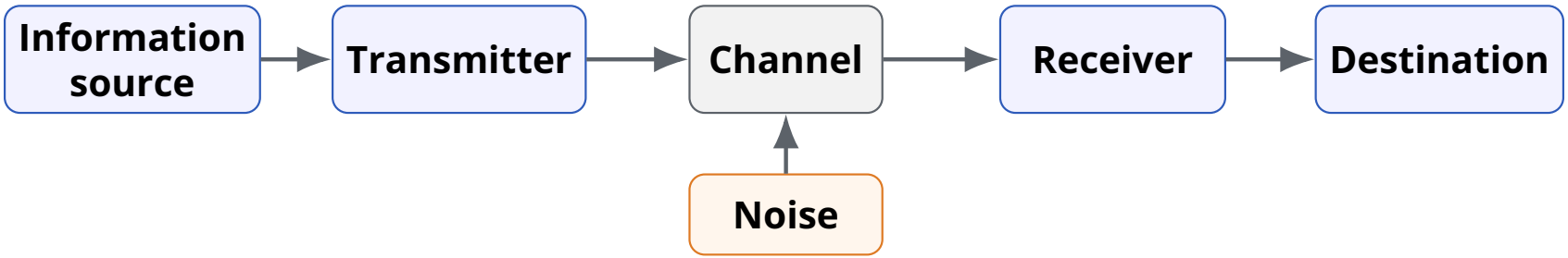
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"These semantic aspects of communication are irrelevant to the engineering problem."

— Shannon (1948)

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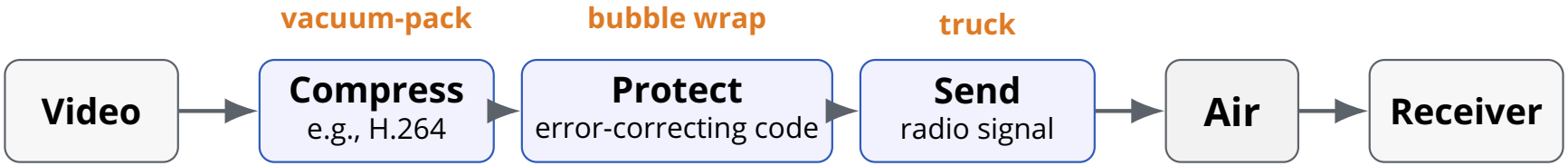
Meaning was left out on purpose:
a bit is a bit, whatever it encodes.

C. E. Shannon, "A Mathematical Theory of Communication," Bell Syst. Tech. J., 1948

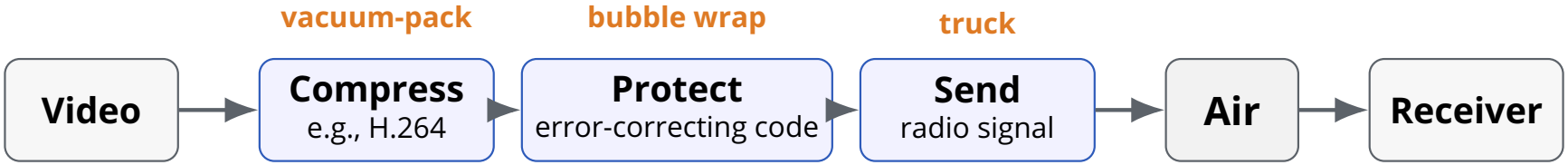
Today’s pipeline: like shipping a suitcase



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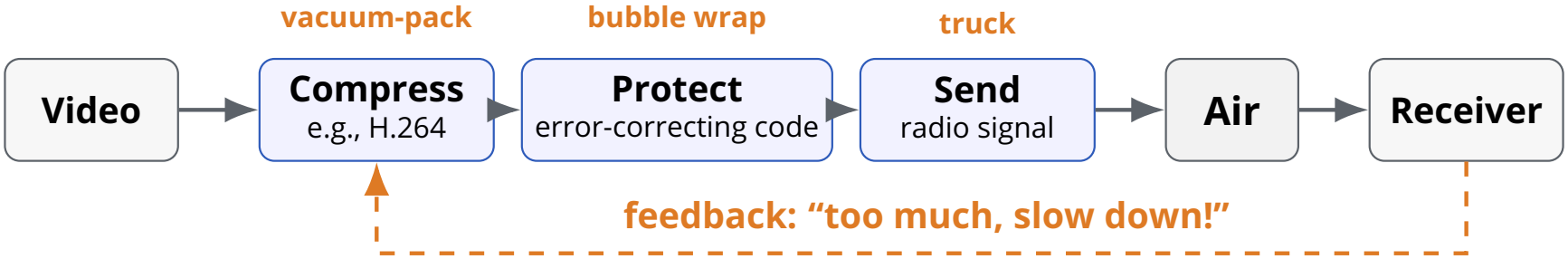


Today’s pipeline: like shipping a suitcase



The truck never opens the suitcase.

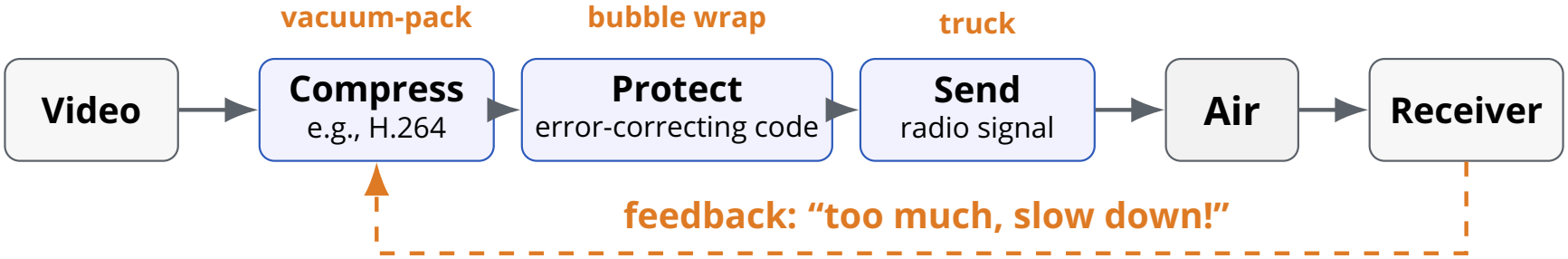
Today’s pipeline: like shipping a suitcase



The truck never opens the suitcase.

The receiver reports back, and the sender adjusts the video quality — like “you’re breaking up” on a phone call.

Today's pipeline: like shipping a suitcase



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Every block has one job:
deliver these bits exactly.

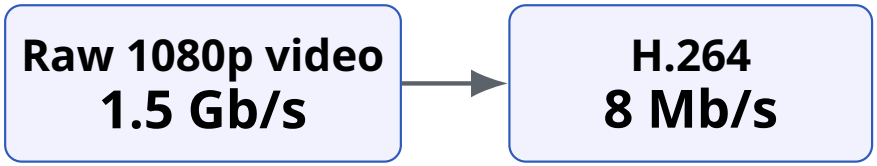
One video stream, by the numbers

Raw 1080p video
1.5 Gb/s

30 frames per second

YouTube recommended upload encoding settings; IEEE Std 802.11; orange box = teaching example

One video stream, by the numbers

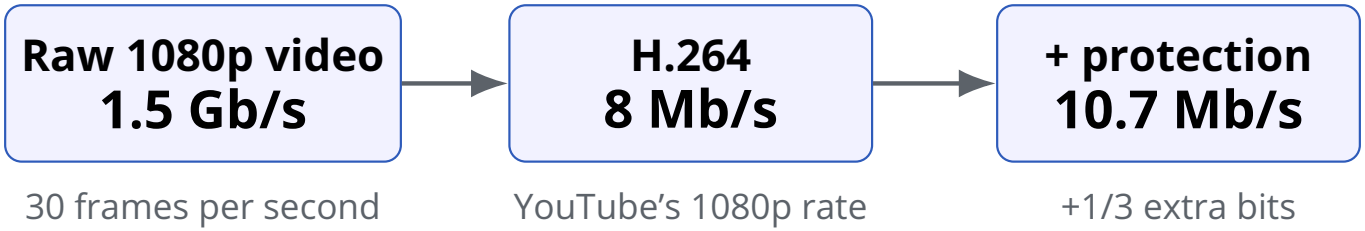


30 frames per second

YouTube’s 1080p rate

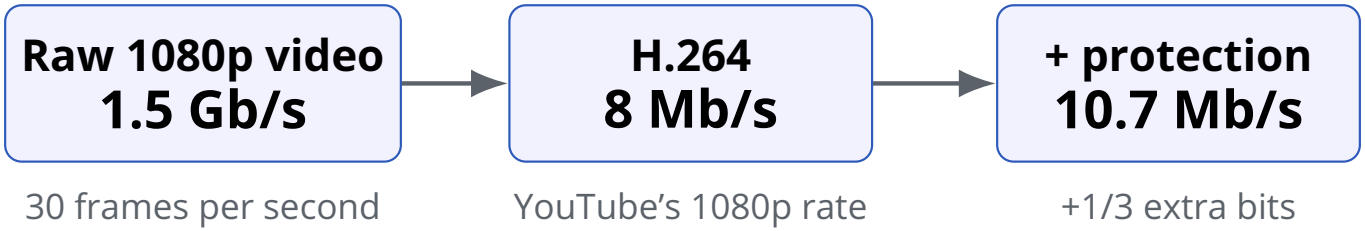
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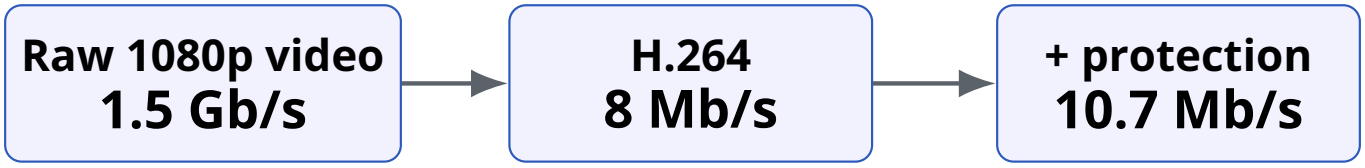
One video stream, by the numbers



But B only needs "2 people": ≈240 b/s

YouTube recommended upload encoding settings; IEEE Std 802.11; orange box = teaching example

One video stream, by the numbers



But B only needs "2 people": ≈240 b/s

Q: What does the receiver check at the end?

A: Only whether every bit is right. One wrong bit, and it is sent again. Nobody asks what the video shows.

YouTube recommended upload encoding settings; IEEE Std 802.11; orange box = teaching example

Why this design won

Divide and conquer

Compress and protect separately: **nothing lost**

(long messages, one link)

Shannon (1948); Cover & Thomas, Elements of Information Theory, 2nd ed., Thm. 7.13.1

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A speed limit

Every channel has a **maximum reliable rate**
$C = B \log_2(1 + \mathrm{SNR})$

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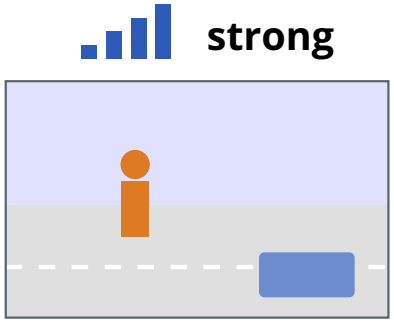
A speed limit

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Bits became a universal interface:
any app over any network.

Shannon (1948); Cover & Thomas, Elements of Information Theory, 2nd ed., Thm. 7.13.1

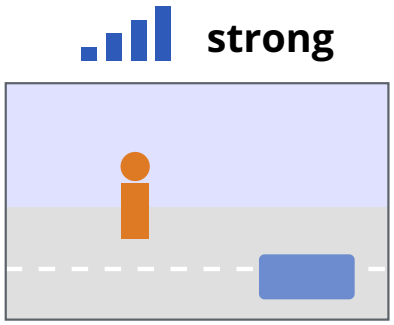
Where it struggles



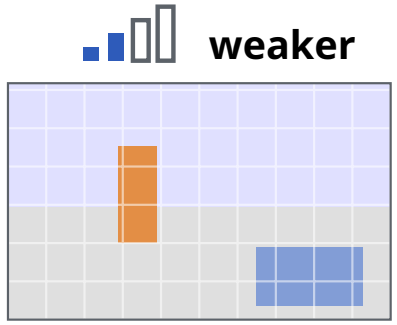


Feedback delay on wireless links: Z. Meng et al., “Zhuge,” ACM SIGCOMM 2022

Where it struggles



sharp

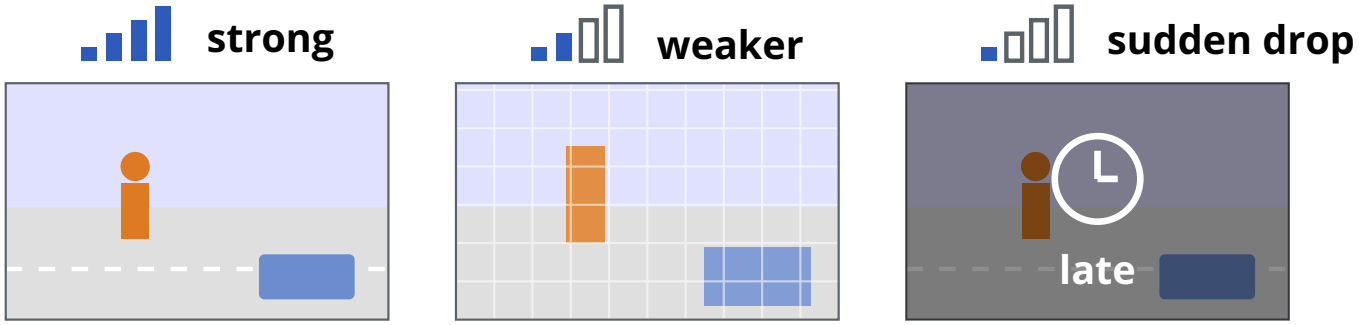


lower quality
feedback cut the bitrate



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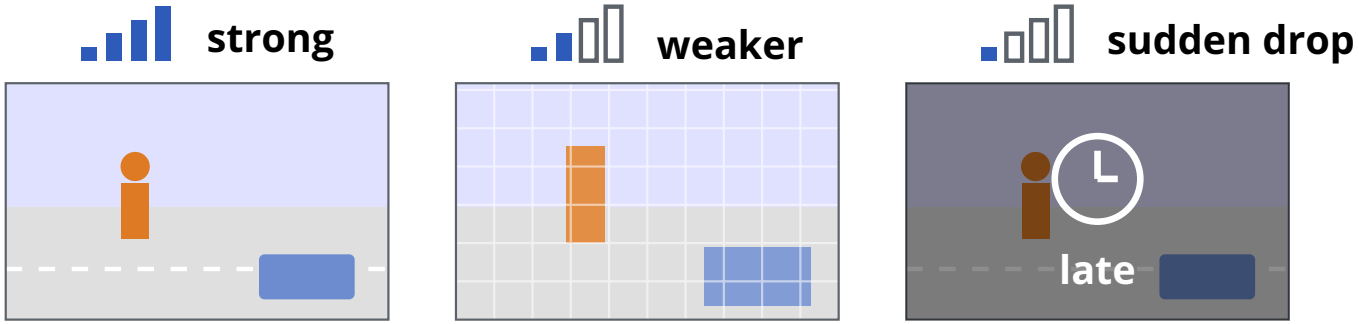
sharp

lower quality
feedback cut the bitrate

late, may freeze
feedback not there yet

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Where it struggles



sharp

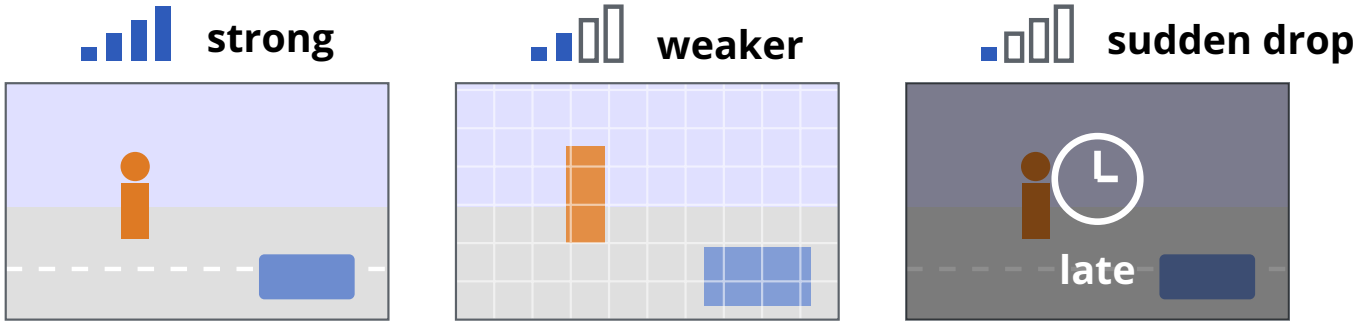
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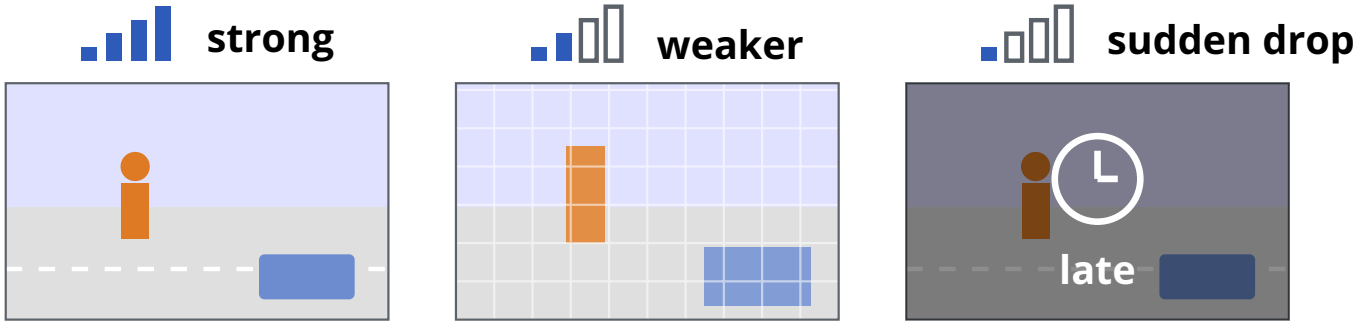
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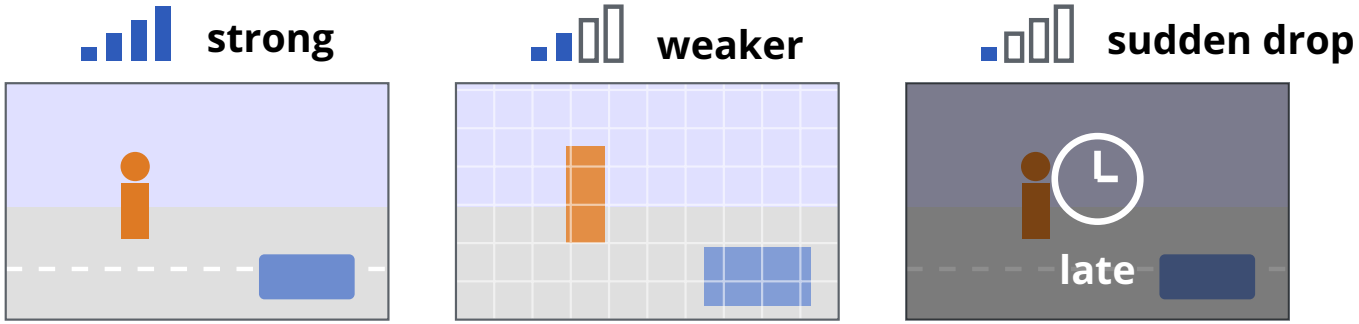
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- **Receiver B only needs an answer**

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- **Receiver B only needs an answer**

Today's video adapts, but after the fact —
and it always sends everything.

Feedback delay on wireless links: Z. Meng et al., "Zhuge," ACM SIGCOMM 2022

Weaver 1949: three levels

A camera streams a street crossing

Level A
Did it arrive intact?

every frame arrives

W. Weaver, “Recent Contributions to the Mathematical Theory of Communication,” in Shannon and Weaver, The Mathematical Theory of Communication, Univ. of Illinois Press, 1949, Sec. 1.2 (paraphrased)

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Level B
Was it understood?

“a pedestrian is crossing”

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Level C Did it lead to the right action?	*the car brakes in time*
Level B Was it understood?	*“a pedestrian is crossing”*
Level A Did it arrive intact?	*every frame arrives*

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Version 2	**Level C** Did it lead to the right action?	*the car brakes in time*
Version 1	**Level B** Was it understood?	*“a pedestrian is crossing”*
Today	**Level A** Did it arrive intact?	*every frame arrives*

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Version 1	**Level B** Was it understood?	*“a pedestrian is crossing”*
Today	**Level A** Did it arrive intact?	*every frame arrives*

Semantic communication aims at Level B or C,
not just Level A.

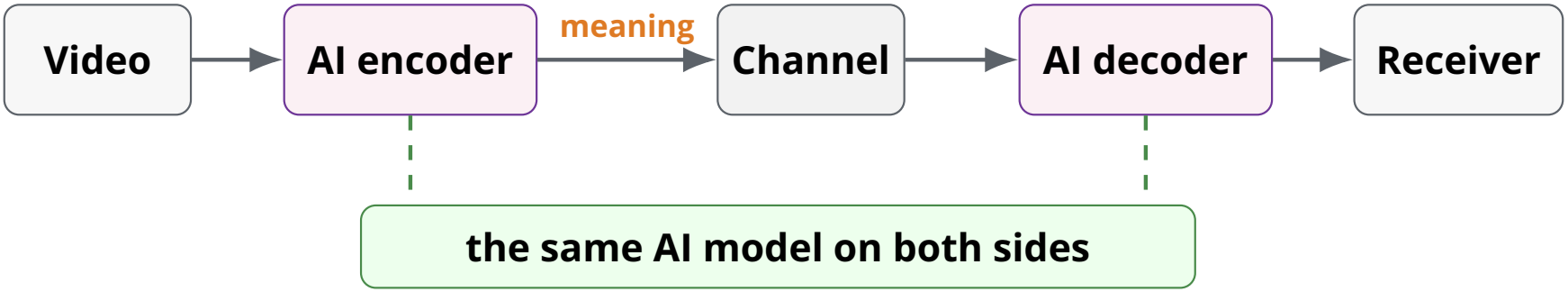
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A semantic communication system



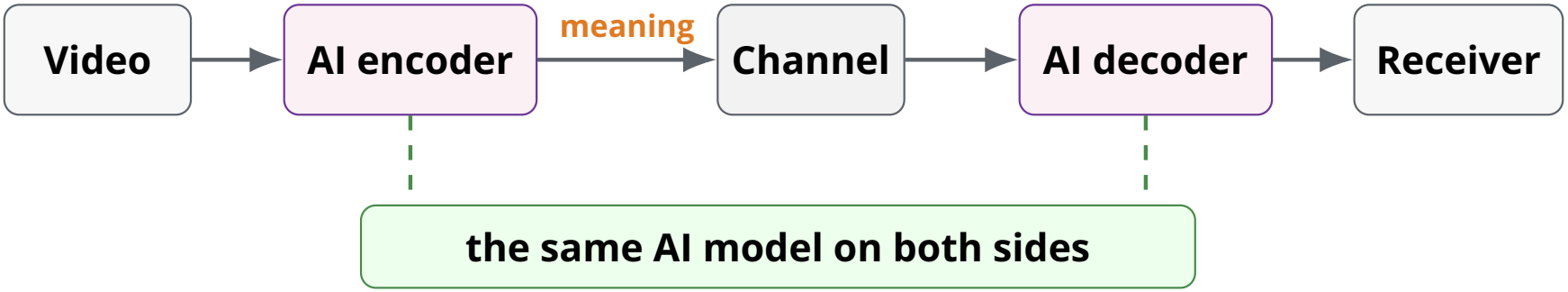
D. Gündüz et al., “Beyond Transmitting Bits,” IEEE JSAC, 2023

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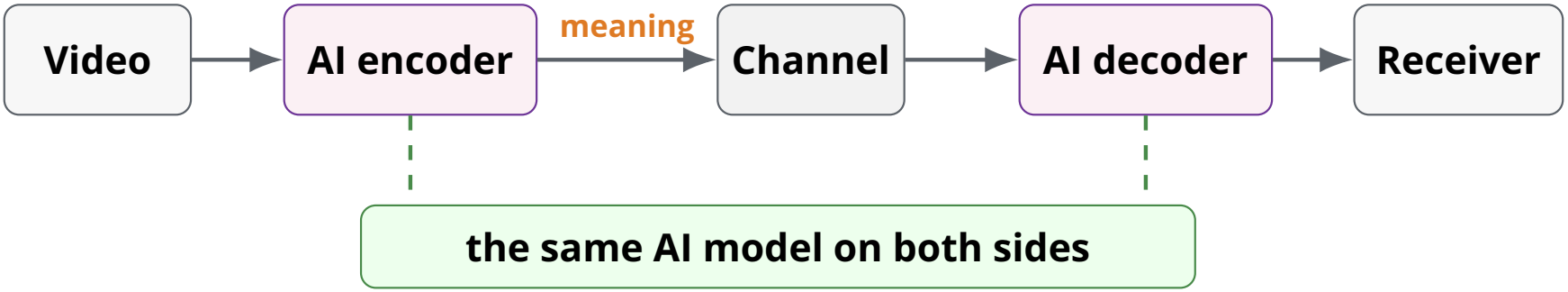
A semantic communication system



“Meet at the usual place” works
only between friends who share context.

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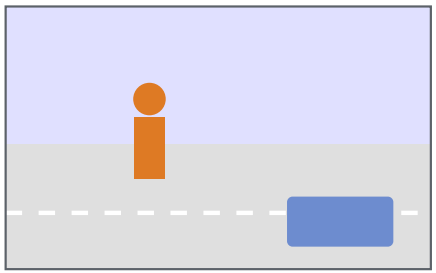
The shared AI model plays the role of shared context.

D. Gündüz et al., "Beyond Transmitting Bits," IEEE JSAC, 2023

Same loss, two outcomes

Each received frame lost the **same** amount of data.

Live video can’t wait for every resend: late data counts as lost.

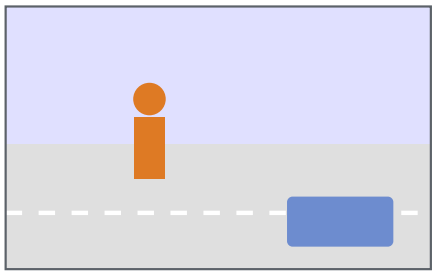


Sent

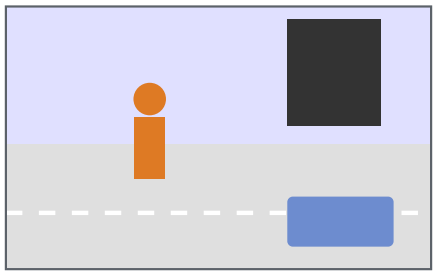
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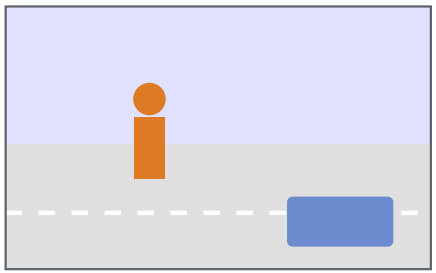
Received 1

lost: sky — fine

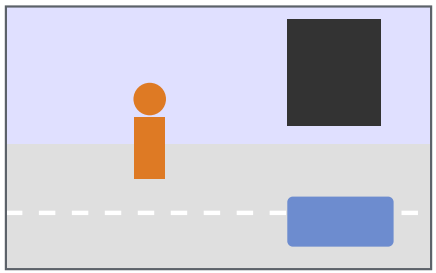
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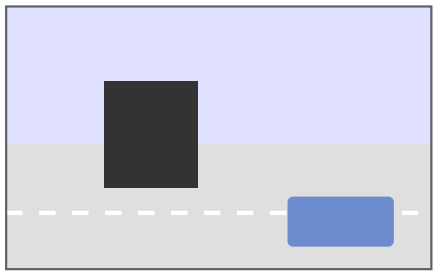


Sent



Received 1

lost: sky — fine



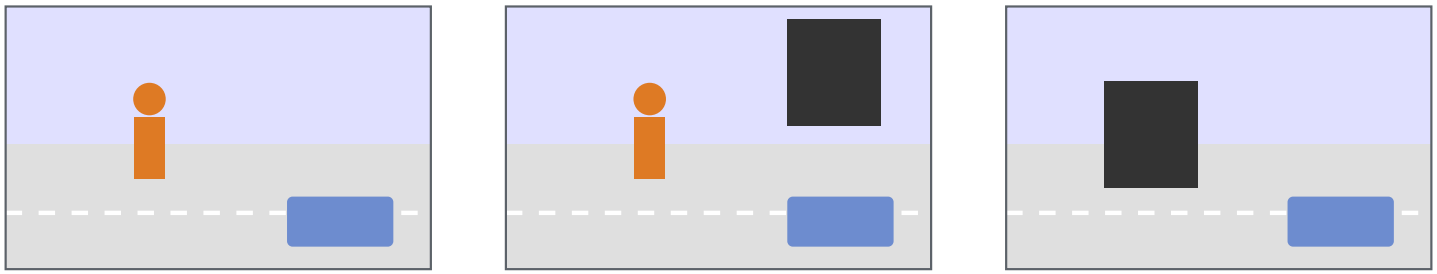
Received 2

lost: pedestrian!

Same loss, two outcomes

Each received frame lost the **same** amount of data.

Live video can't wait for every resend: late data counts as lost.



Sent

Received 1
lost: sky — fine

Received 2
lost: pedestrian!

"Meaning delivered" means…

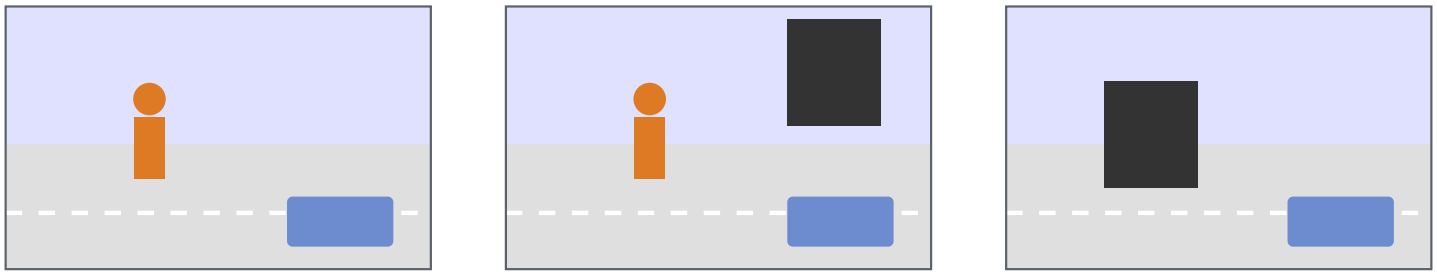
Version 1: rebuild it
Does the video look right?

Version 2: do the task
Was the pedestrian found?

Same loss, two outcomes

Each received frame lost the **same** amount of data.

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Sent

Received 1
lost: sky — fine

Received 2
lost: pedestrian!

"Meaning delivered" means...

Version 1: rebuild it
Does the video look right?

Version 2: do the task
Was the pedestrian found?

Same amount of lost data, different damage.

Less bandwidth

- Rebuilt video:
  up to 50% less (DVST)

Wang et al., JSAC 2023; Cheng et al., SenSys 2024; Guo et al., JSAC 2025; Bourtsoulatze et al., TCCN 2019; Cheng et al., NSDI 2024

What we gain

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- 3D video call:
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Survives a bad network

- Signal drops:
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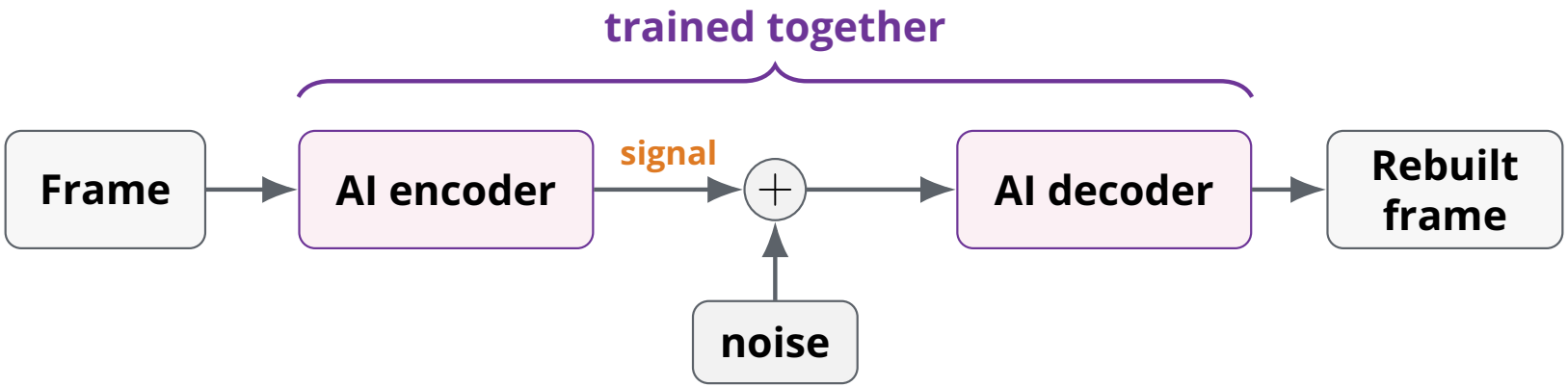
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Best cases reported by each paper, each against its own baseline.

Send less, and break less
when the network misbehaves.

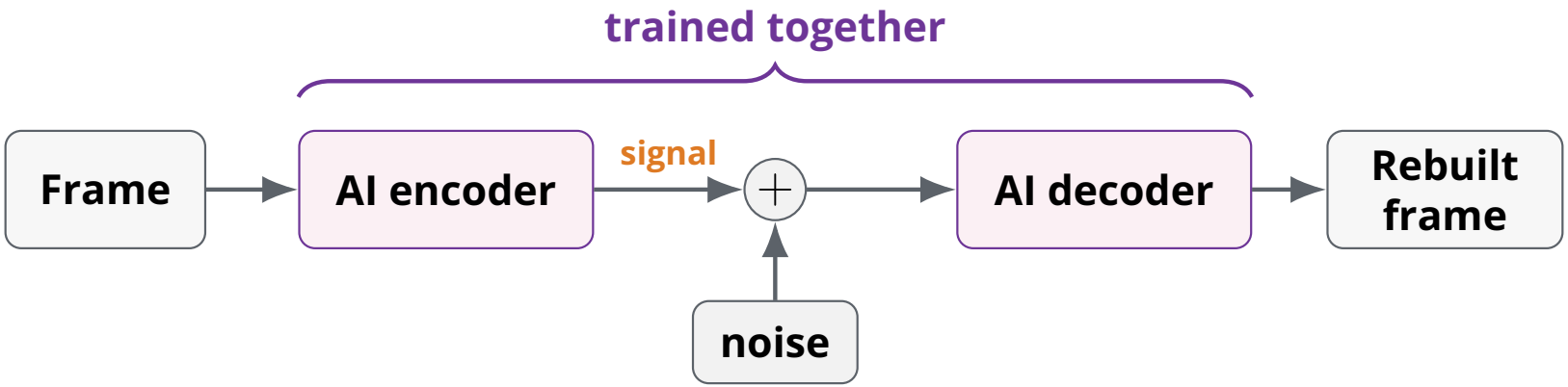
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How Version 1 works: practice with noise



First shown for images (“deep JSCC”): Bourtsoulatze, Kurka & Gündüz, IEEE TCCN, 2019

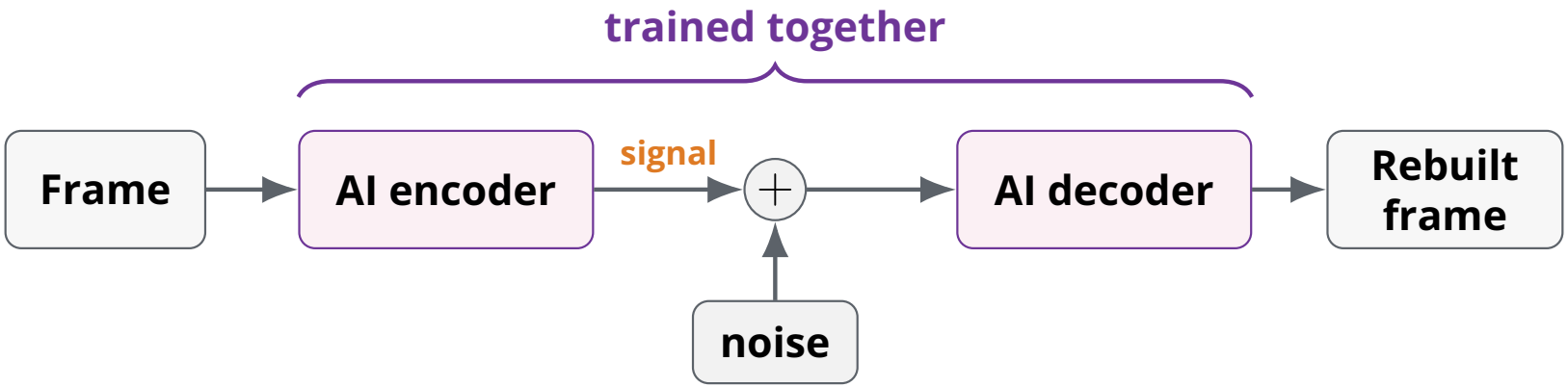
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- One AI replaces “compress, then protect”

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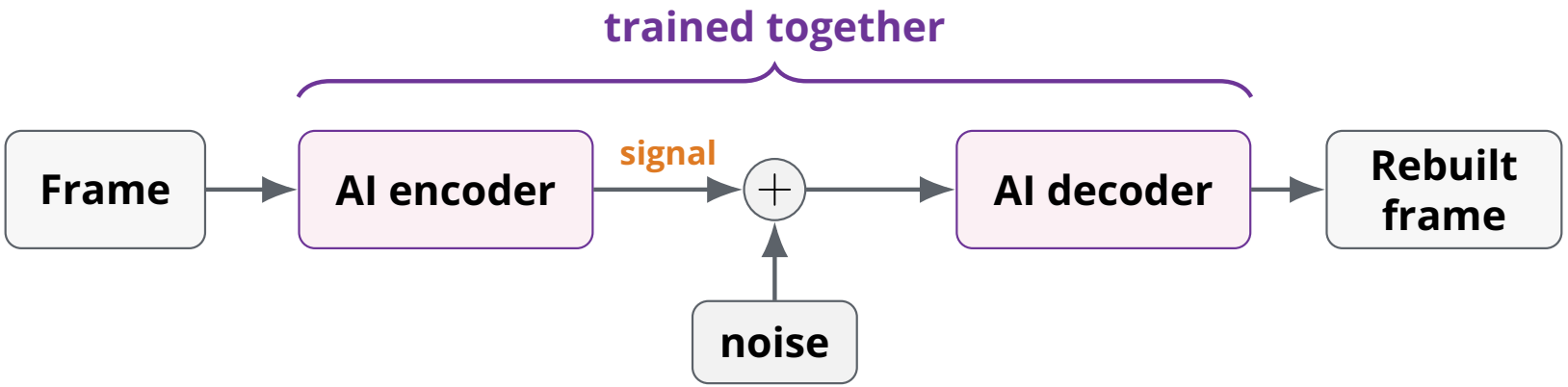
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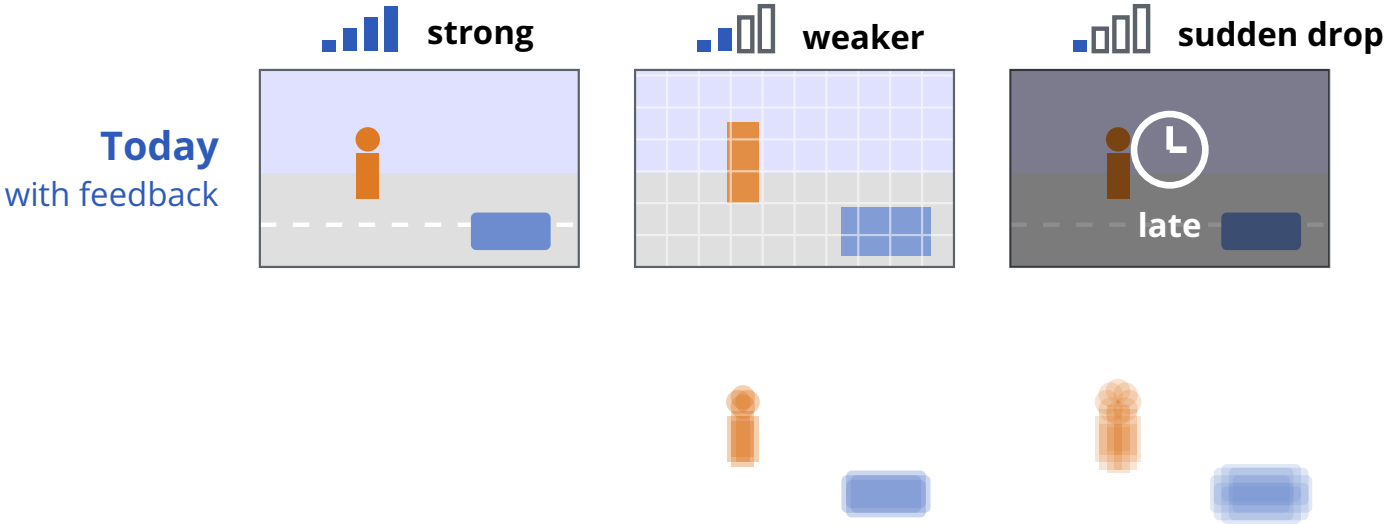


- One AI replaces “compress, then protect”
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Like two partners practicing a code
until it survives a noisy phone line.

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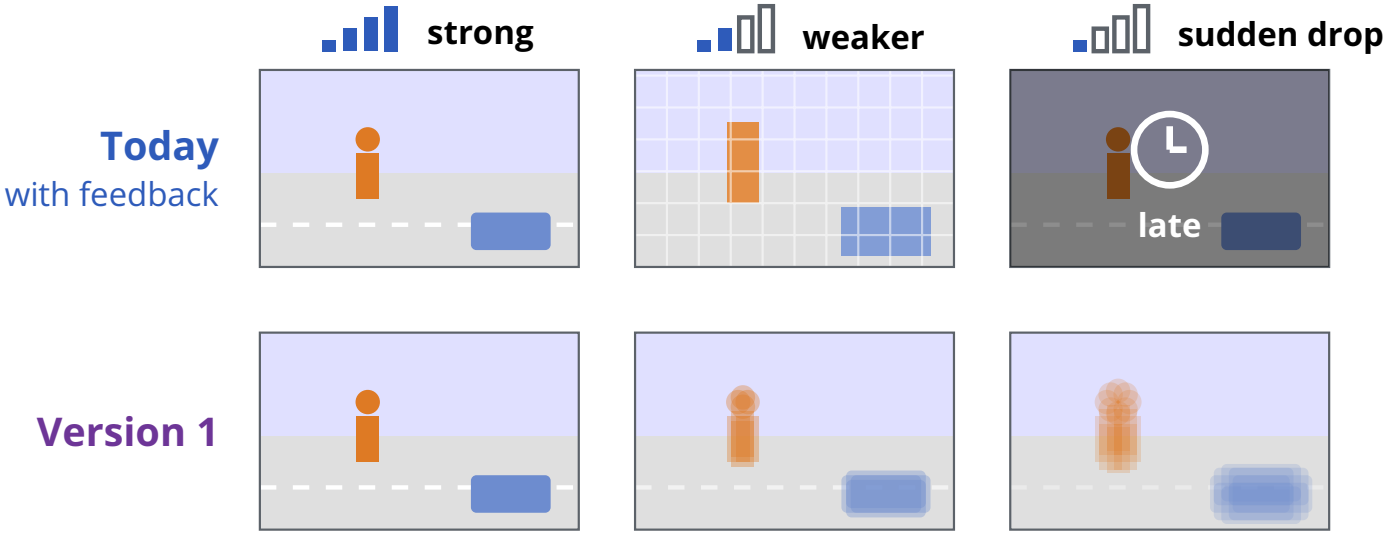
Weak signal: no waiting for feedback



Sketch, not output from the paper.

Bourtsoulatze, Kurka, Gündüz, IEEE TCCN, 2019 (compared with digital schemes without feedback)

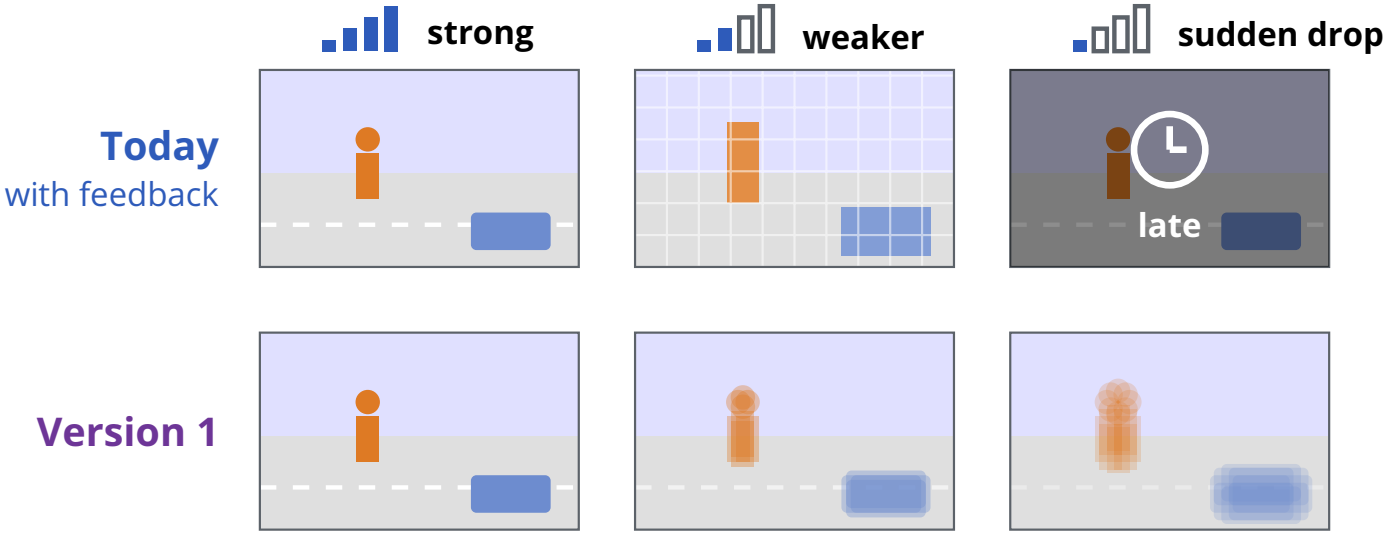
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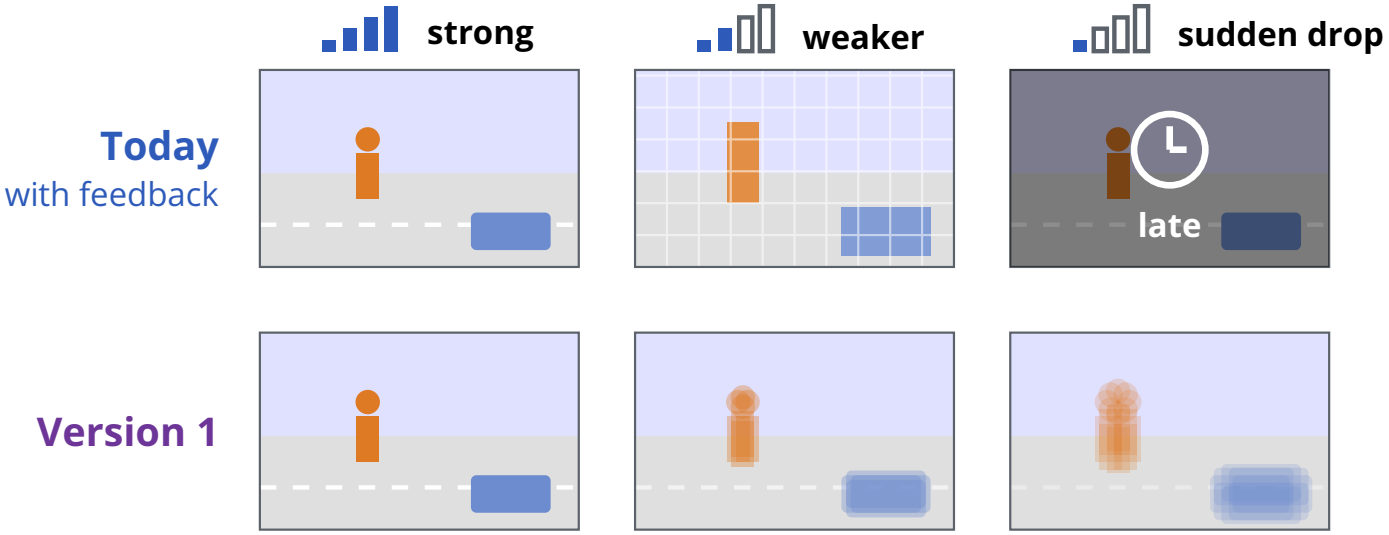


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- Picture follows the signal **at once** — no feedback round trip

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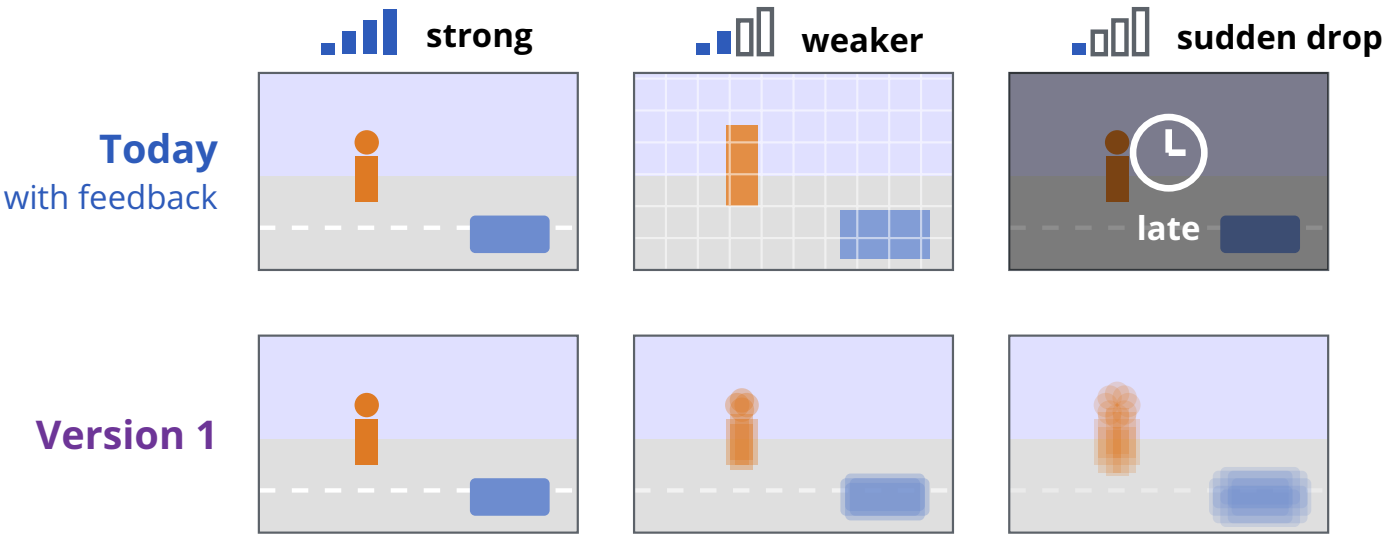


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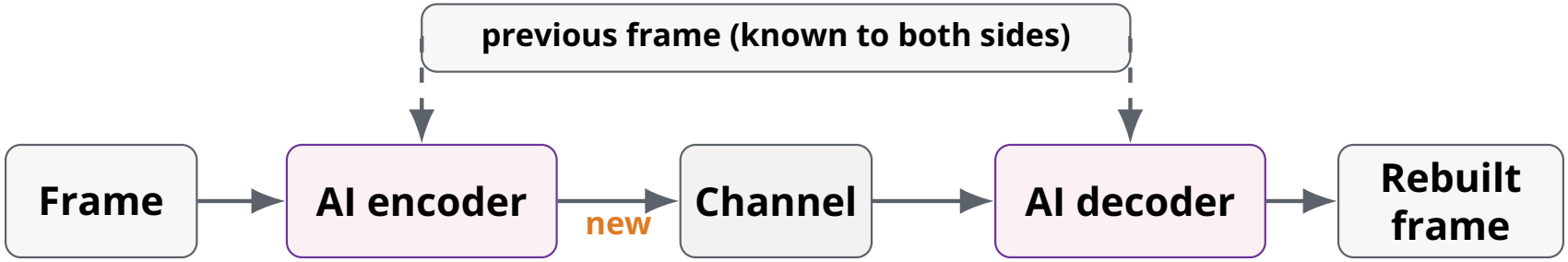
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Its strength is reacting instantly,
not “always better.”

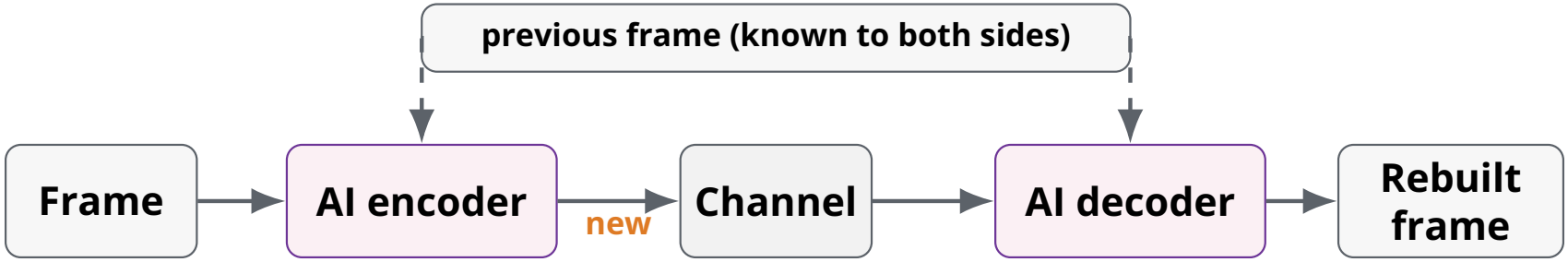
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Version 1 for video: send what's new



Example system: DVST, S. Wang et al., "Wireless Deep Video Semantic Transmission," IEEE JSAC, 2023

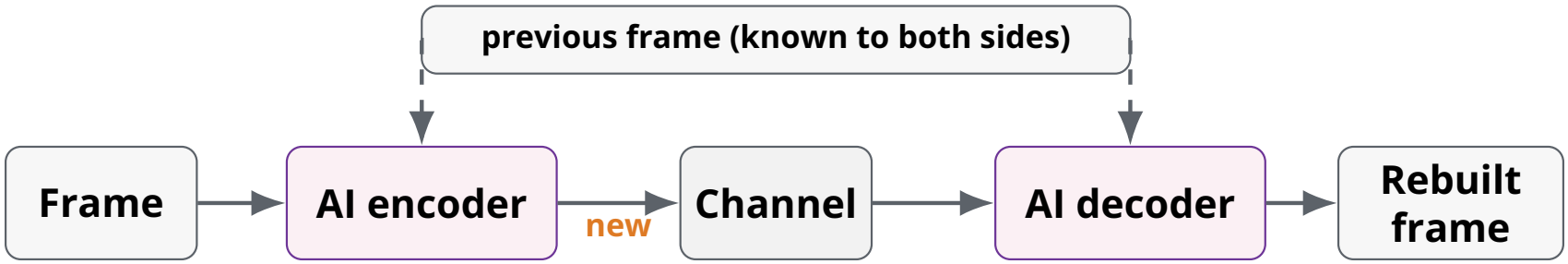
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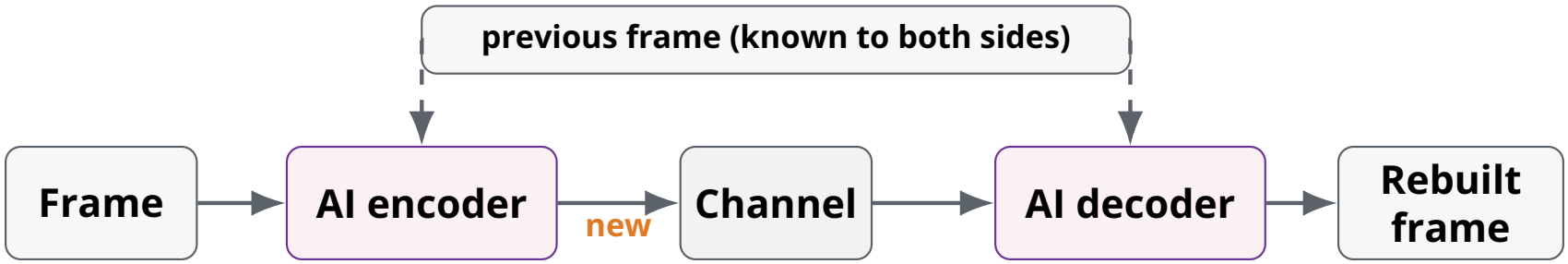
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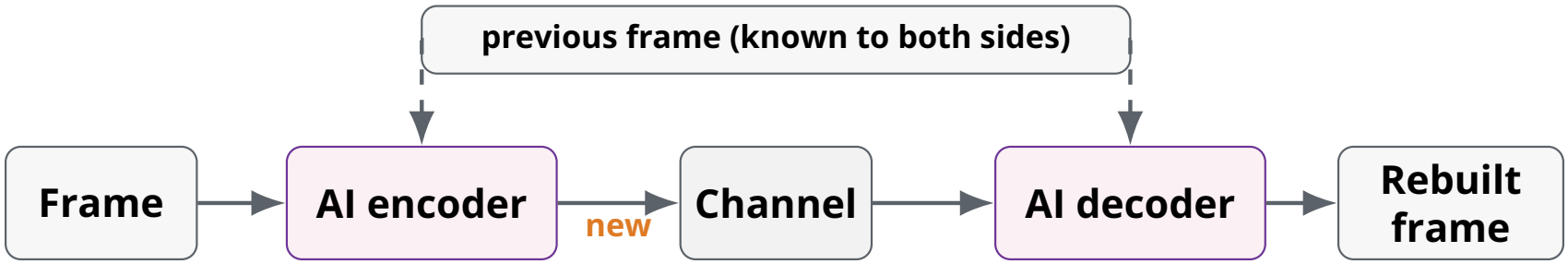
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- **Up to 50% less bandwidth** than H.264/H.265 + error coding

Same video, up to half the bandwidth —
but every frame is still rebuilt.

Example system: DVST, S. Wang et al., "Wireless Deep Video Semantic Transmission," IEEE JSAC, 2023

The price of rebuilding

1. **Radios only speak a fixed “alphabet”**
   learned signals must be squeezed into it

Point 1: T.-Y. Tung et al., “DeepJSCC-Q: Constellation Constrained Deep JSCC,” IEEE JSAIT, 2022

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Smooth quality is paid for with
compatibility, shared models, and trust.

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Version 2: when the receiver is a machine

Receiver B only asks: **“any pedestrians?”**

1. **Send the video** (several Mb/s): the server does all the work

Kang et al., Neurosurgeon, ASPLOS 2017; Shao & Zhang, BottleNet++, ICC Workshops 2020

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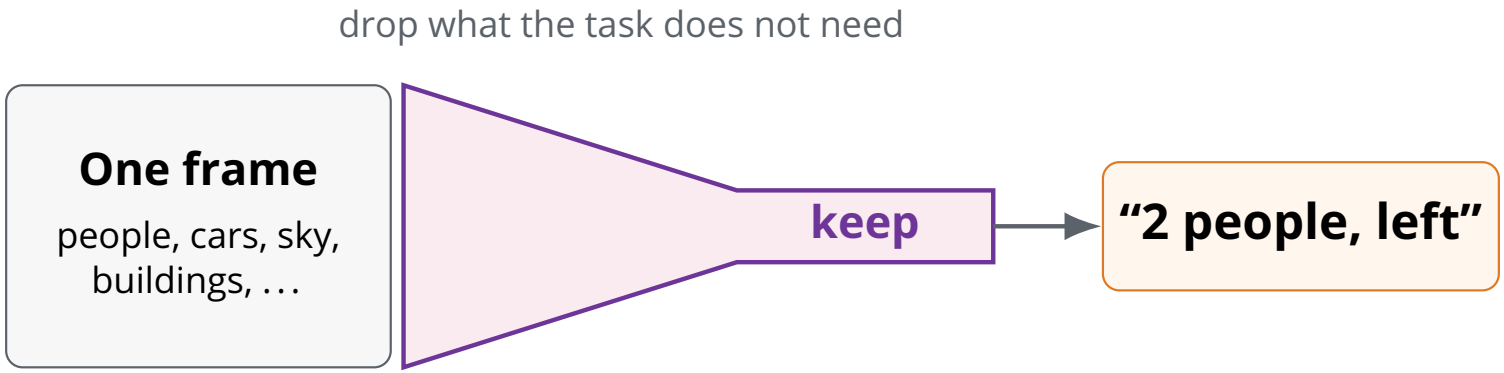
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With clues, no frame is ever rebuilt —
only the decision matters.

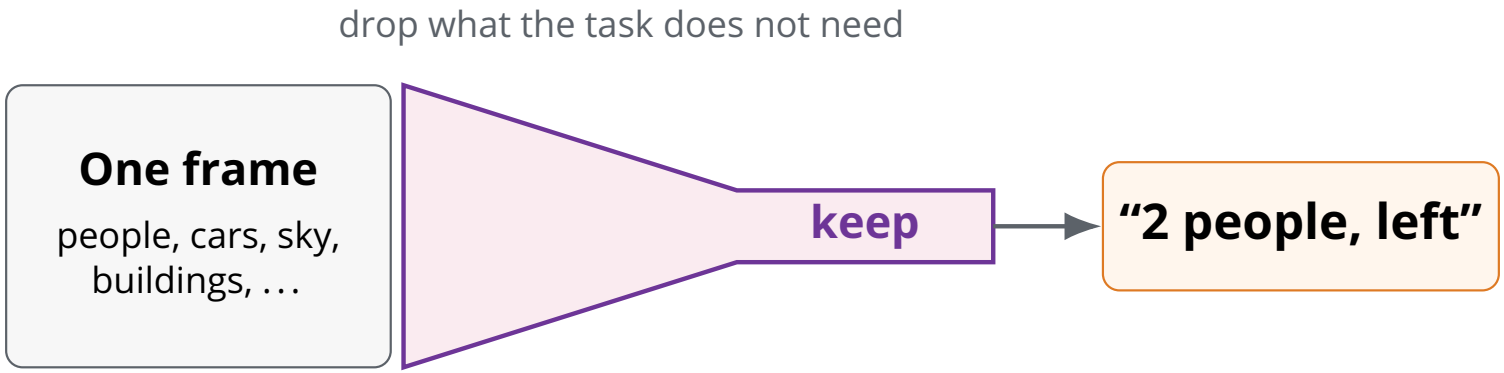
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How Version 2 works: keep what the task needs



The “information bottleneck” approach: J. Shao, Y. Mao, J. Zhang, IEEE JSAC, 2022

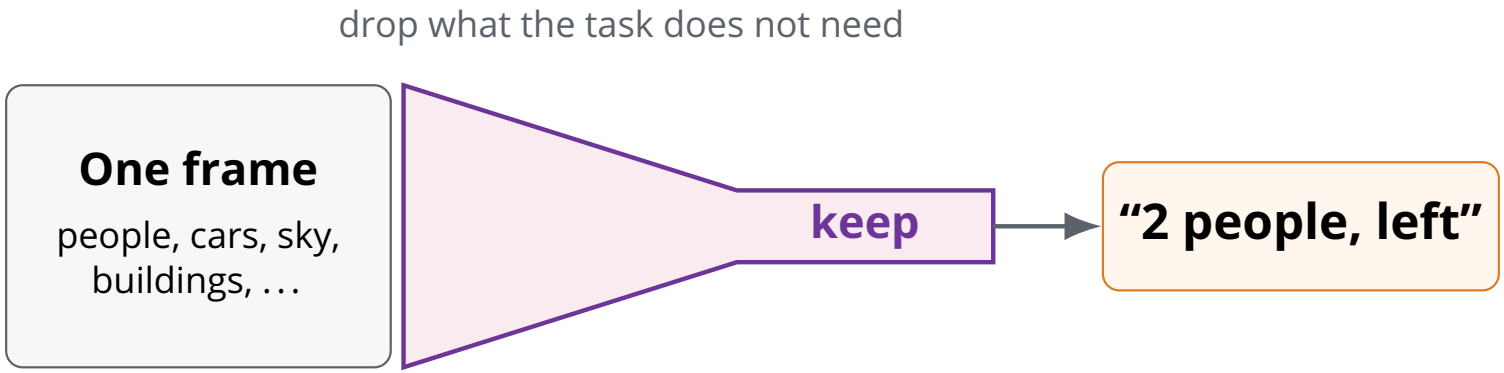
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- Finding people? Keep their shapes, drop the sky

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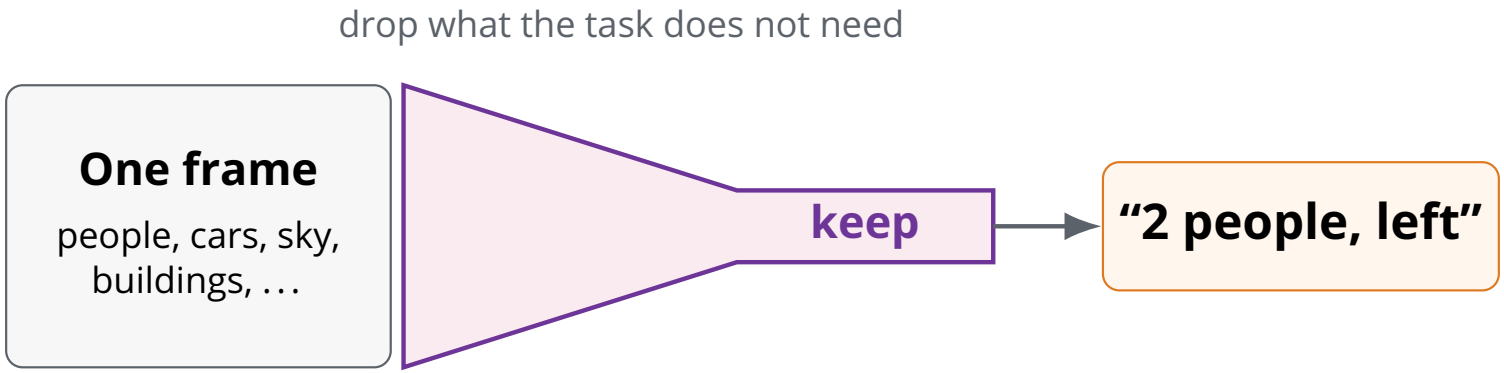
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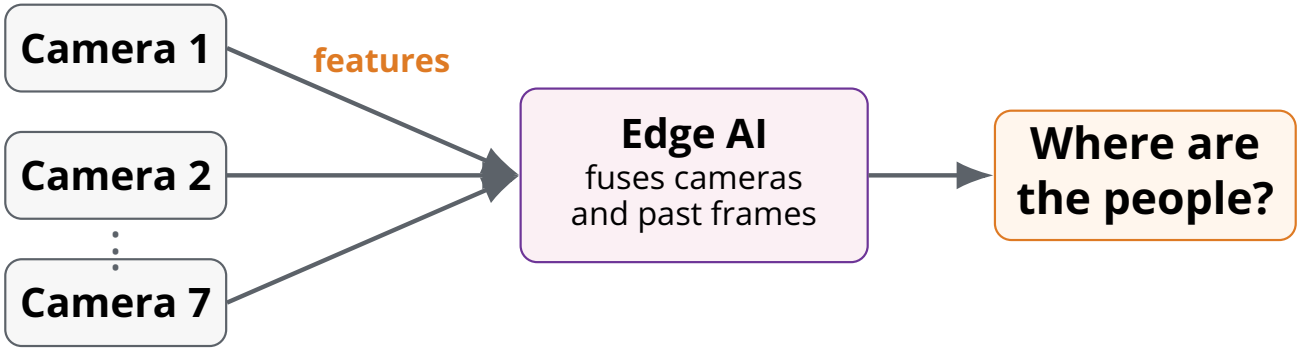


- Finding people? Keep their shapes, drop the sky
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Send “2 people, left” instead of every pixel.

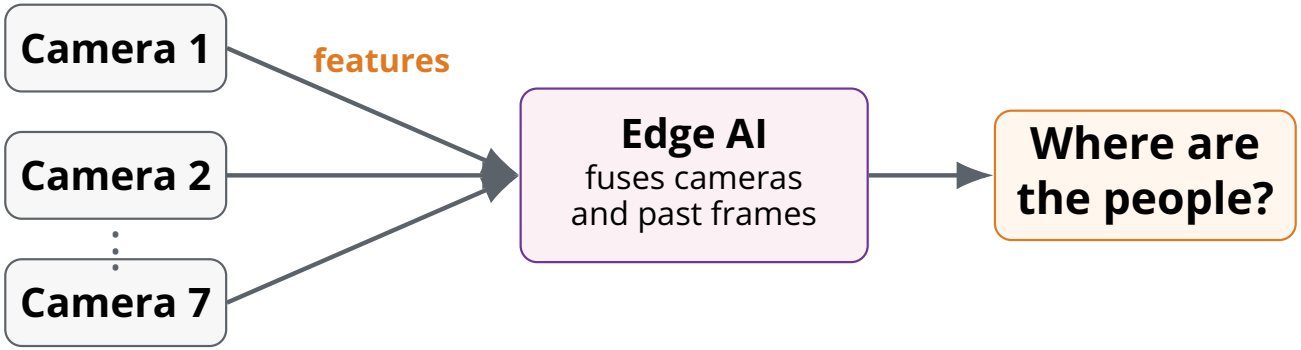
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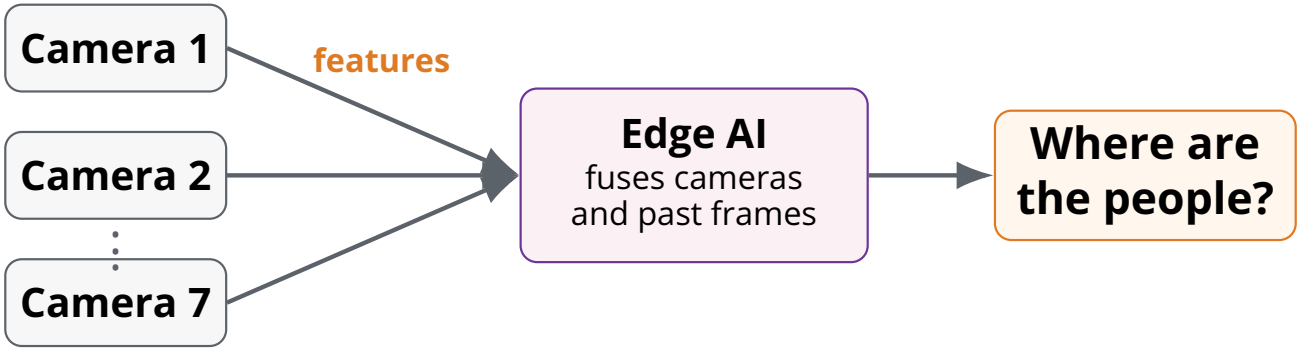
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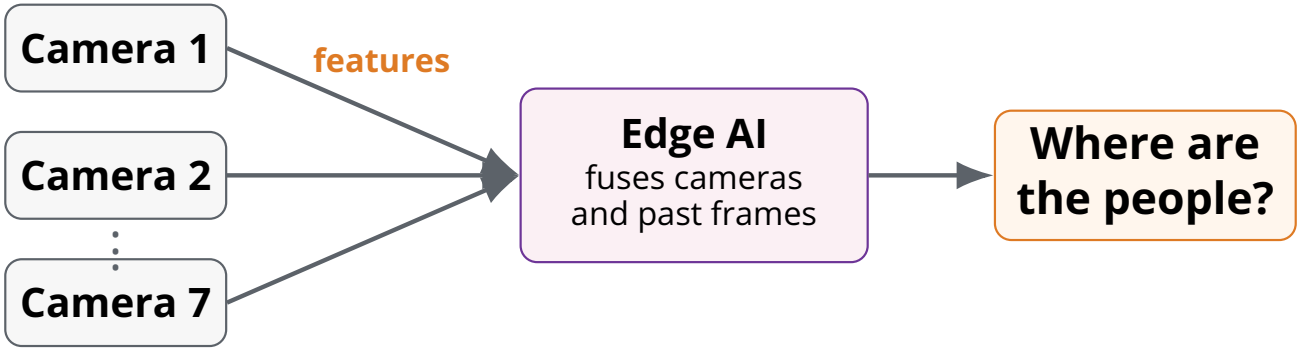
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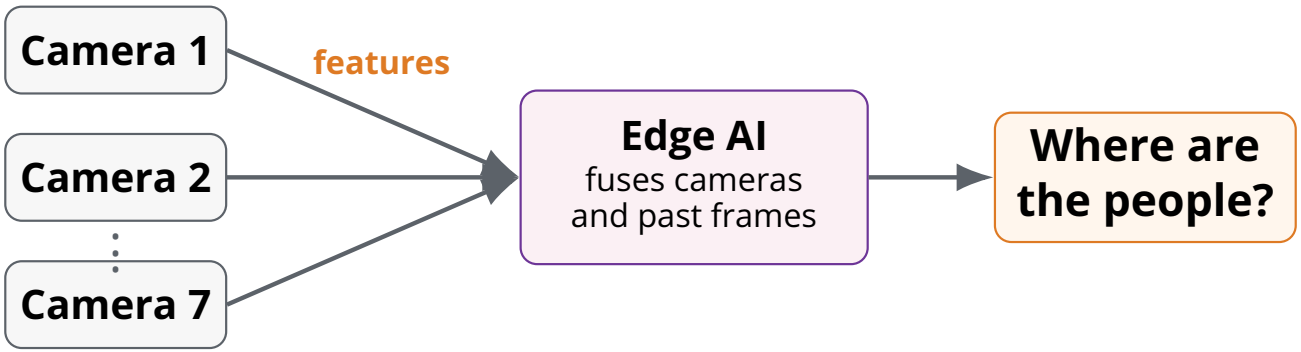
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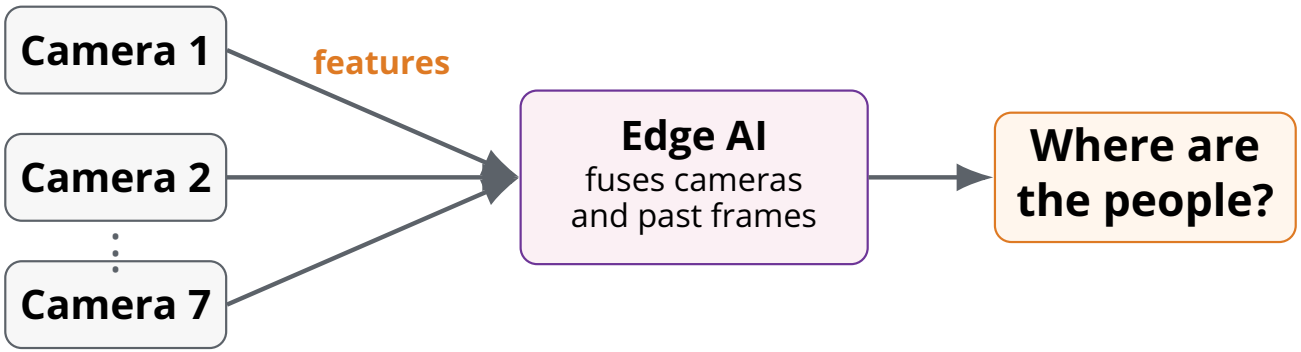
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For machines, skip the pixels.

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Send only when the answer changes

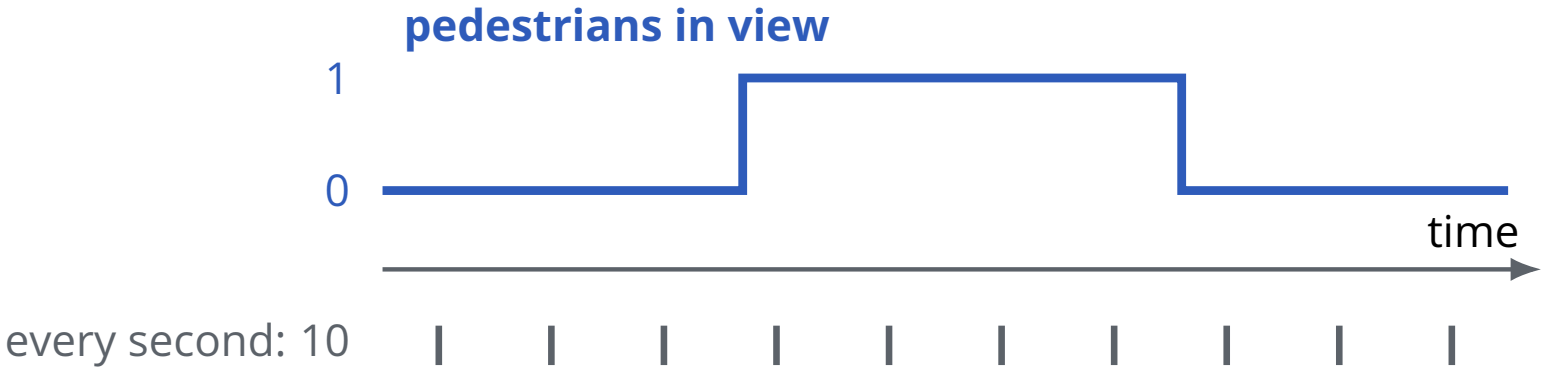
An extra option, not a rule: many task-oriented systems still send something every frame.



Yates et al., AoI survey, IEEE JSAC, 2021; Kountouris & Pappas, IEEE Commun. Mag., 2021

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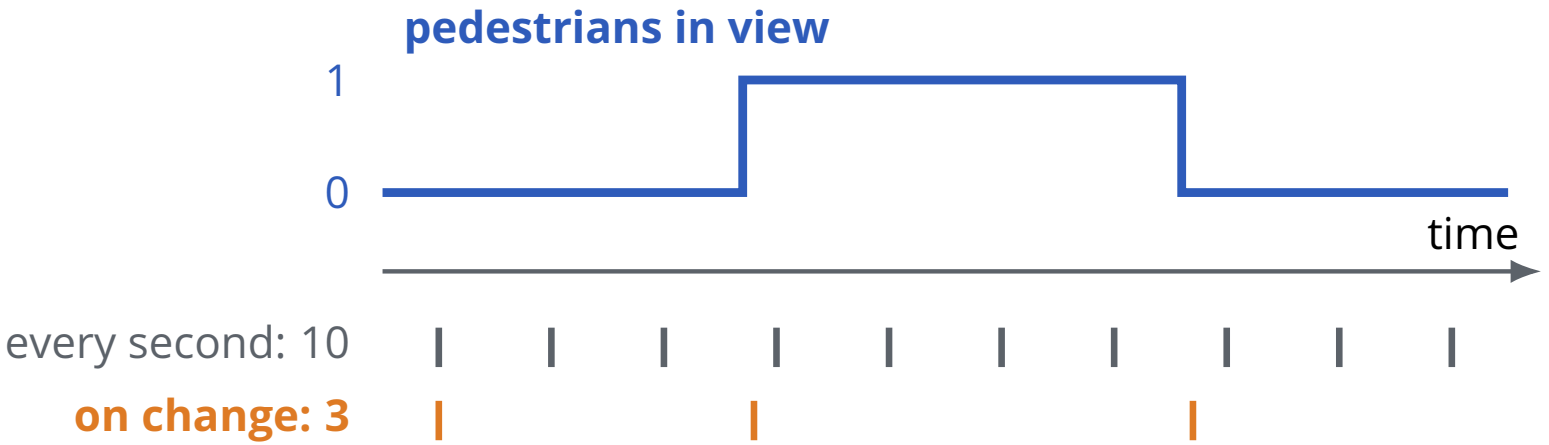
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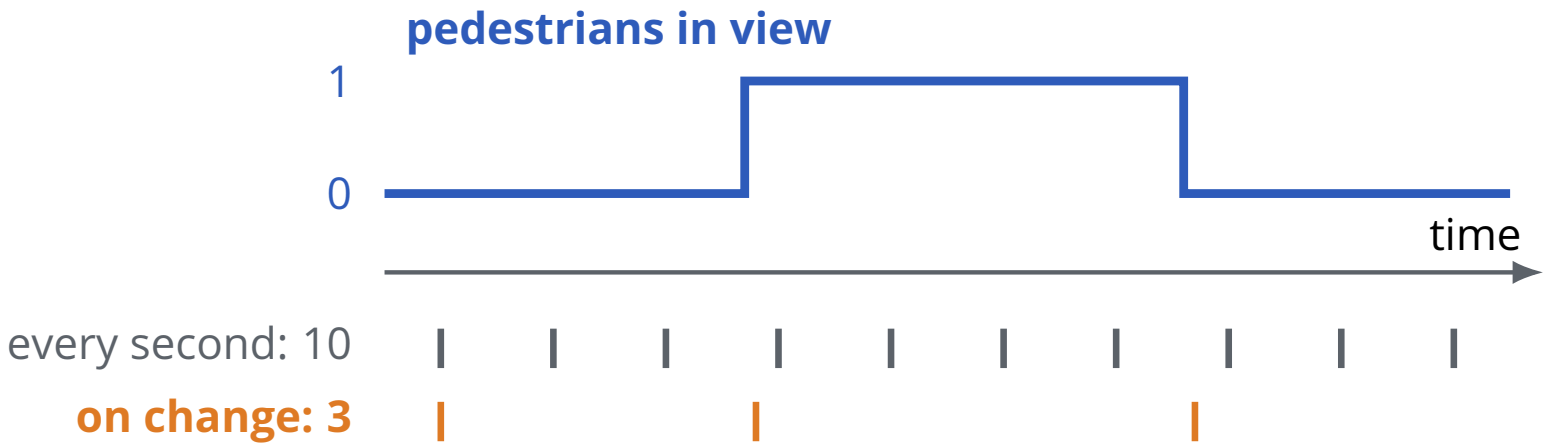
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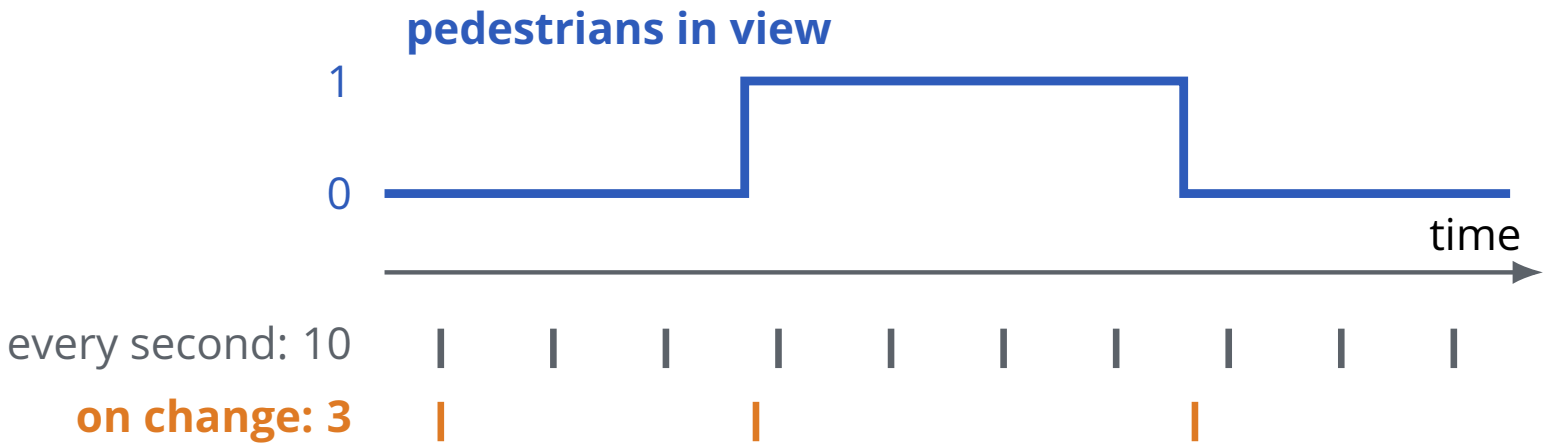


- **Freshness:** how old is my latest update? (“Age of Information”)

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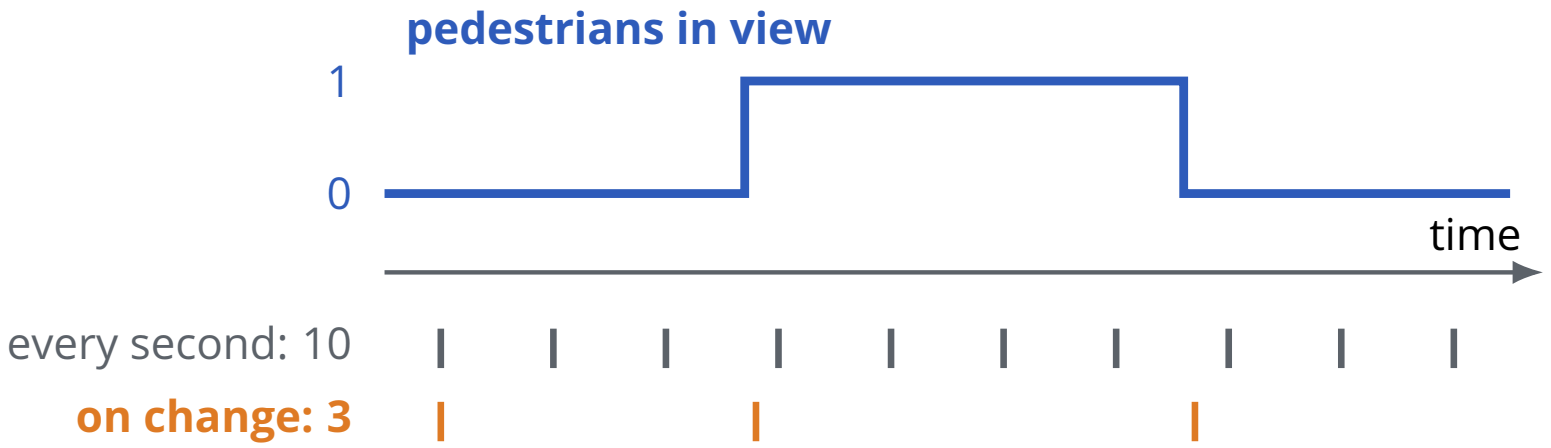


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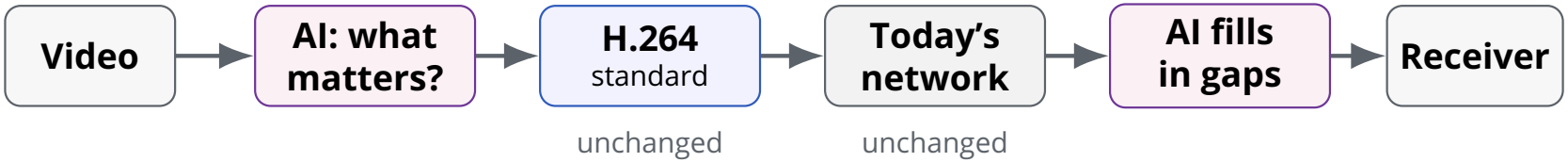


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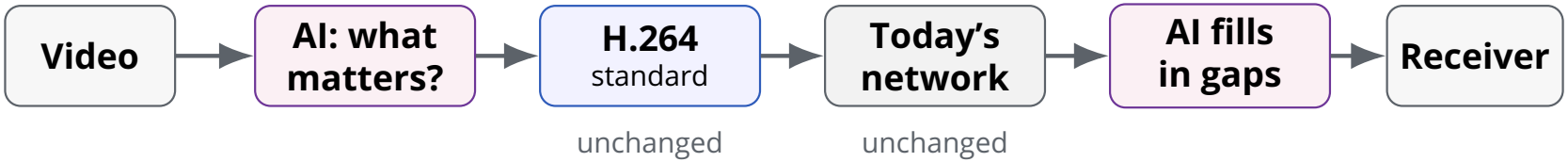
Another lever: decide **when** to send, too.

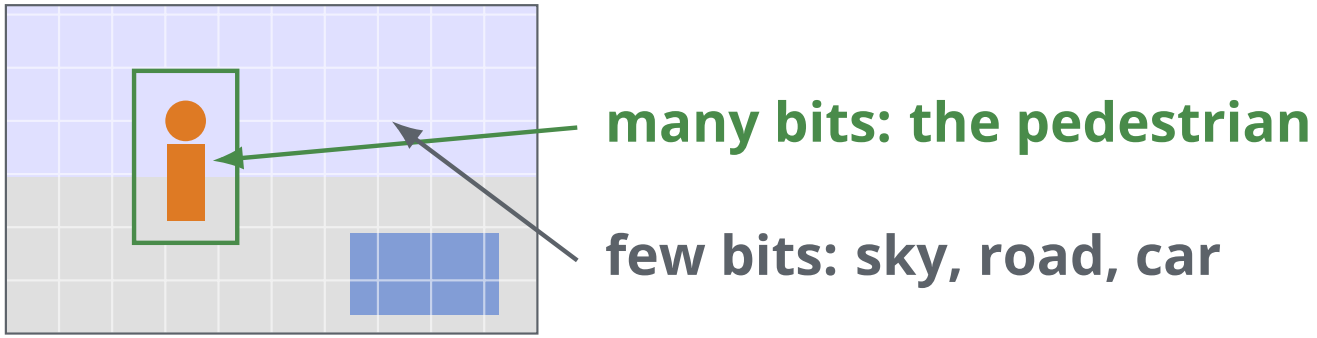
Yates et al., AoI survey, IEEE JSAC, 2021; Kountouris & Pappas, IEEE Commun. Mag., 2021

Version 3: keep the pipe, spend bits by meaning

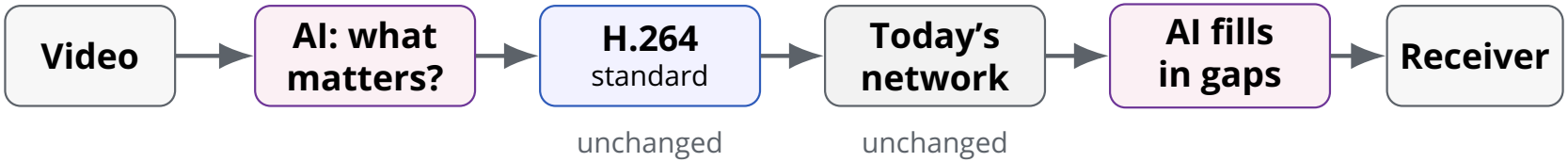


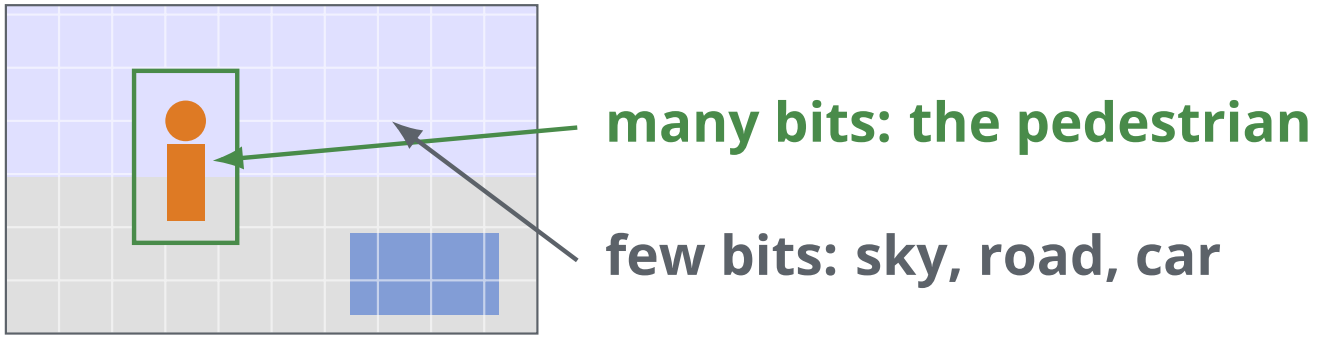
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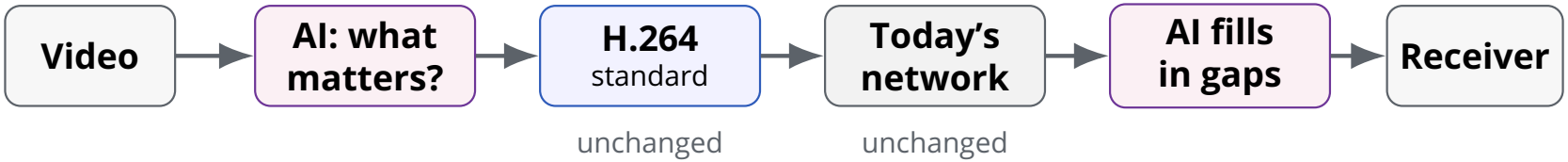
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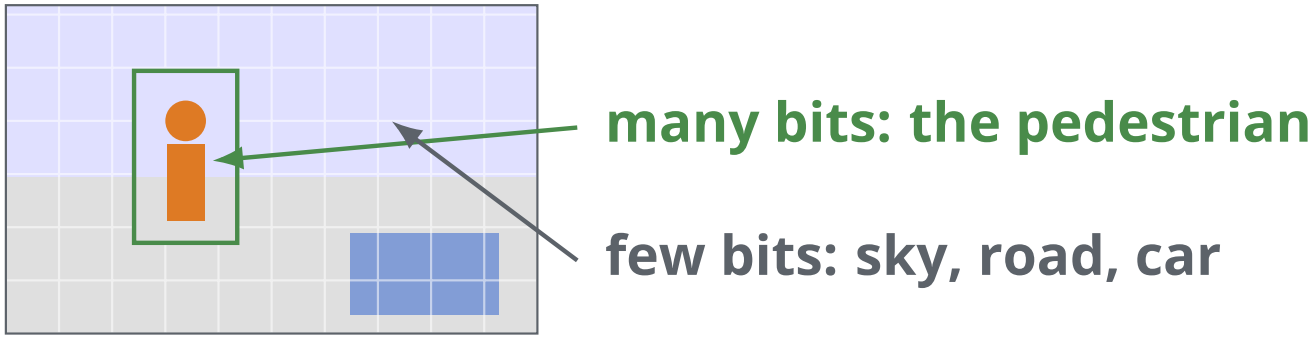




Nothing new for radios, codecs, or networks.

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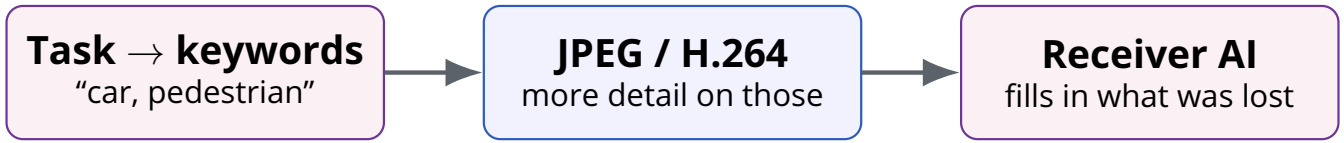


Nothing new for radios, codecs, or networks.

Same pipe, smarter spending —
the easiest version to deploy today.

Version 3 in practice

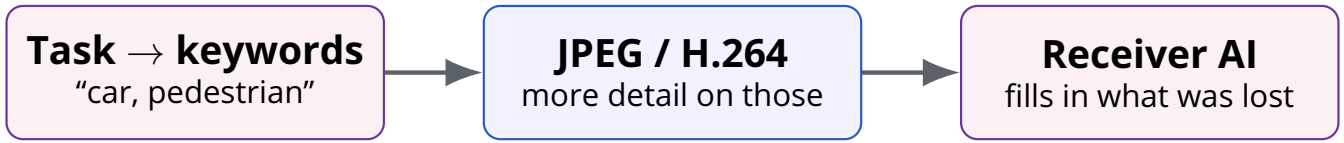
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X. Chen et al., “From Bits to Tokens: Knowledge-Driven Generative Communication,” USENIX NSDI, 2026

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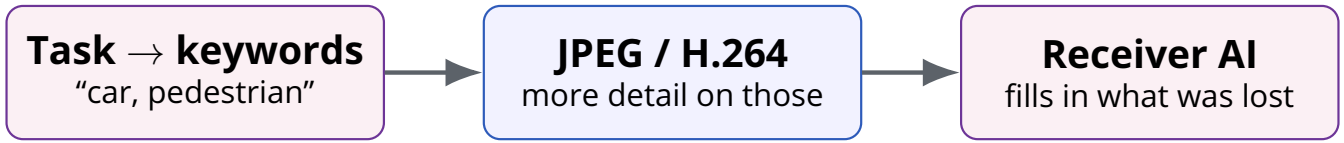


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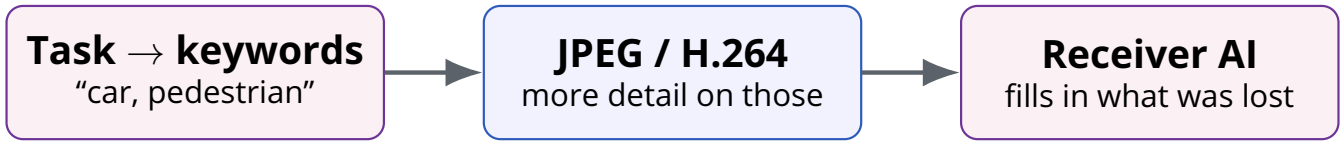


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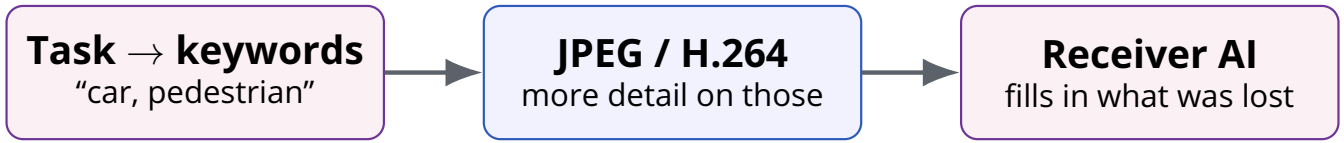


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Works with today's codecs and networks —
but still sends a picture, and needs AI on both ends.

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Four ways to send video

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No single winner:
choose by what the receiver needs.

Why it is not in your phone yet

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3GPP TR 38.843 (Rel-18); MPEG press releases, meetings 153–155 (2026)

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Promising for weak signals and clear goals;
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Back to the start: do A and B need the same data?